



Voluntary Carbon Standard 2007.1
Validation Report

19 November 2007

Validation Report:

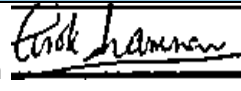
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Validation Report for "Grid connected electricity generation from renewable sources: Uzundere I 63.0 MW Hydroelectric Power Plant Project, Turkey"	Ashok Mammen 
Client:	Project Title:
Karadeniz Elektrik Uretim San. A.S.	"Grid connected electricity generation from renewable sources: Uzundere I 63.0 MW Hydroelectric Power Plant Project, Turkey"
Summary:	
Bureau Veritas Certification has made the validation of "Grid connected electricity generation from renewable sources: Uzundere I 63.0 MW Hydroelectric Power Plant Project, Turkey". The project activity involves the installation of a 63.0 MW hydro electric power plant (HEPP) located on the Uzundere River in the Black Sea Region of Turkey, within the province of Rize. The project comes under Type-I Renewable Energy Project as per Appendix B of the procedures for CDM project activities. The expected annual emission reduction is 87,318 tons/year. The crediting period of the project activity is 10 years, renewable twice for a total of 30 years. The validation scope is defined as an independent and objective review of the VCS project description, the project's baseline study, monitoring plan and other relevant documents, and consisted of the following five phases: i) desk review of the project design and the baseline and monitoring plan; ii) Review of the baseline methodology by the specialist and the validator; iii) Review of the investment analysis by the specialist and the validator iv) follow-up interviews and on site visit; v) resolution of outstanding issues and the issuance of the final validation report and opinion. The overall validation, from Contract Review to Validation Report & Opinion, was conducted using Bureau Veritas Certification internal procedures. The first output of the validation process is a list of Clarification and Corrective Actions Requests (CL and CAR), presented in Annex II in the Validation Protocol. Taking into account this output, the project proponent revised its VCS project description. In summary, it is Bureau Veritas Certification's opinion that the project correctly applies the baseline and monitoring methodology ACM0002, version 11 and meets the relevant VCS 2007.1 requirements and local legislation.	
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Abbreviations List:

HEPP	: Hydro Electric Power Plant
CL	: Clarification
CAR	: Corrective Action Request
MP	: Monitoring Plan
VCS	: Voluntary Carbon Standard
VCU	: Voluntary Carbon Unit
VER	: Voluntary Emissions Reductions / Verified Emissions Reductions
PD	: Project Description
EMRA	: Electricity Market Regulatory Authority (EPDK)
EPDK	: Elektrik Piyasasi Duzenleme Kurulu (EMRA)
TEIAS	: Turkiye Elektrik Iletim A.S. (Turkish Electricity Transmission Company)
DSI	: Devlet Su Isleri (State Hydraulic Works)
PP	: Project Participant
CDM	: Clean Development Mechanism
HES	: Hidro Elektrik Santrali (Hydro Electric Power Plant)
FSR	: Feasibility Study Report

1 INTRODUCTION

This report summarizes the findings of the validation of the "Grid connected electricity generation from renewable sources: Uzundere I 63.0 MW Hydroelectric Power Plant Project, Turkey" project, performed on the basis of VCS 2007.1 criteria, as well as criteria given to provide for consistent project operations, monitoring and reporting.

1.1 Objective

The validation serves as project design verification and is a requirement of all voluntary emission reduction projects. The validation is an independent third party assessment of the project design. In particular, the project's baseline, the monitoring plan (MP), and the project's compliance with relevant VCS criteria which are validated in order to confirm that the project design, as documented, is sound and reasonable, and meets the stated requirements and identified criteria. Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of Voluntary Carbon Units (VCUs).

1.2 Scope and Criteria

The validation scope is defined as an independent and objective review of the VCS project description, the project's baseline study and monitoring plan, the project's investment analysis and other relevant documents. The information in these documents is reviewed against the methodologies and tools that are given under the Clean Development Mechanism of Kyoto Protocol, VCS 2007.1 rules and associated interpretations. The validation is not meant to provide any consulting towards the client. However, stated requests for clarifications and/or corrective actions may provide input for improvement of the project design.

1.3 VCS Project Description

The project activity involves the installation of a 63.0 MW hydroelectric power plant (HEPP) on the Uzundere River in the Black Sea Region of Turkey, within the province of Rize. The project comes under Type I – Renewable Energy Project as per Appendix B of the procedures for CDM project activities. The main purpose of the project is to generate approximately 156,205 MWh/year of electricity to supply the national grid using a renewable resource and tapping the significant hydropower potential in the region. The project activity reduces greenhouse gas (GHG) emissions that would have otherwise occurred in the absence of the project activity by avoiding electricity generation from fossil fuel sources. The average annual emission reductions of the proposed Project are estimated to be 87,318 tonnes of CO₂e (tCO₂e). The crediting period of the project activity is 10 years, renewable twice for a total of 30 years.

1.4 Level of assurance

The Level of assurance of the validation report is defined as high.

2 METHODOLOGY

The overall validation, from contract review to validation report & opinion, was conducted using Bureau Veritas Certification internal procedures.

In order to ensure transparency, a validation protocol was customized for the project. The protocol shows, in a transparent manner, criteria (requirements), means of verification and the results from validating the identified criteria. The validation protocol serves the following purposes:

- It organizes, details and clarifies the requirements of a VCS project is expected to meet;
- It ensures a transparent validation process where the validator will document how a particular requirement has been validated and the result of the validation.

2.1 Review of Document

The Project Description (PD) submitted by Karadeniz Elektrik Uretim San. A.S. (KARADENIZ) and additional background documents related to the project design and baseline, i.e. relevant Law, VCS Project Description Template, Approved Methodology and tools under the CDM of Kyoto Protocol, Clarifications (CL) and Corrective Action Requests (CAR) on validation requirements were reviewed.

To address Bureau Veritas Certification corrective action requests and clarification requests, KARADENIZ revised the PD several times, PD version 07 dated 14.03.2011 being the last version, which was submitted to the validation team on 16.03.2011. After further clarifications all corrective action requests and clarification requests were closed in 16.03.2011. The validation of the Project resulted in 10 Corrective Action Requests and 28 Clarification Requests.

All documents were reviewed under the requirements of the CDM Methodologies as these methodologies are approved under the VCS.

2.2 Follow-up Interviews

Between March 2010 and December 2010 Bureau Veritas Certification performed interviews with KARADENIZ and local stakeholders regarding the Grid emission factor, Investment Analysis, project specific questions and Clarification and Corrective Actions Requests documented in the validation protocol. A list of the persons interviewed is given under section 5.2 of this report. After the discussions, the validation team closed on 16.03.2011 all documented CL and CAR in the validation protocol.

2.3 Resolution of any material discrepancy

During the validation there were no material discrepancies that were identified.

3 VALIDATION FINDINGS

3.1 Project Design

Bureau Veritas Certification recognizes that KARADENIZ, which is the project participant, is helping the country in fulfilling its goals of promoting sustainable development with this project. The project is in line with country specific requirements. In addition, the project highly supports the sustainable economic development in the region.

It was validated during the site visit that the project description given in the VCS PD version 07 is accurate. It was also crosschecked with the project description given in the feasibility study report.

Uzundere I HPP is not a debundled component of a larger project activity as defined in the VCS PD version 07. The project activity is neither in existing facilities nor in utilizing existing equipments. During the site visit, it was confirmed that project activity is a greenfield project.

The conclusions of the validation team are as follows:

- The technology used:

Uzundere I HEPP project entails the construction and operation of a 63.0 MW hydroelectric power plant (HEPP) in the Black Sea Region of Turkey, in Uzundere River Basin in the province of Rize, Turkey. The location of the project activity was validated during the site visit.

In the powerhouse, two Pelton type horizontal turbines were installed each with the capacity of 31.6 MW; the total capacities of the turbines are 63 MW. The altitude is 817.59 m at powerhouse location. There are two generators attached to the facility. Generators pursue power factor of 0.9 d/d, a frequency of 50 Hz and an output of 34.5 MVA.

The implementation of the project consists of construction of the following main items:

- Two diversion weirs, where water from the river is diverted into conveyance pipes
- Total length of the conveyance tunnel is 11,324 m
- Power house with Pelton type vertical turbines

The net electricity delivery to the grid by the project is estimated to be 156,205 MWh per annum.

The technology used in the plant is a state-of-the-art technology with extensive automatization. In addition to this, the equipments that are used in the project are all first-hand equipments. There is no new technology, which is expected to replace the plant's technology in the short run, and the expected life of the equipment is around 20 years. The project does not need extensive initial training and maintenance efforts in order to work as presumed during the project period. Necessary trainings to the plant staff was provided by the supplier right after the installation of the electro mechanic equipments in line with the agreement between the project owner and the supplier. The agreement between the supplier and Uzundere I HEPP has been reviewed by the validation team. The related training records shall be verified during the first verification of the project activity.

As a result, validation team approves that the technology used in the project activity is state of the art and all the equipment used in the project are new.

- Project duration, crediting time and project start date

Uzundere I HEPP consists of 2 unit turbines. Each unit has an output capacity of 31.6 MW. The project activity is implemented in 2 phases. On 27/05/2010, the power plant started operation with 1 turbine and on 07/10/2010, the remaining 1 turbine was commissioned. Each commissioning date was validated through the "Provisional Acceptance Certificate" approved by EMRA (Energy Market Regulations Authority).

The carbon crediting period and therefore the monitoring starts when the plant commences electricity generation. Therefore, the crediting period starts on 27/05/2010. The crediting life is a maximum of ten years that may be renewed at most two times.

The expected lifetime of the project is, starting from August 2006, minimum 49 years, which corresponds to the lifetime of the license obtained from

EMRA. However, the equipments may be renewed, as recommended, in years according to the conditions of the equipments. The owner of the company will be able to renew the license, by applying to EMRA in 9 – 12 months before the expiration date of the first license duration.

- Ownership

The generation license belongs to Karadeniz Elektrik Uretim San. A.S.. A copy of the generation license is given in Annex 2 of the PD.

The project did not participate in any other GHG emission reduction program. The validation team approves that there is no double counting involved in this project activity during this validation.

- Eligibility of the project activity under VCS

The project comes under Type I – Renewable Energy Project as per Appendix B of the procedures for CDM project activities. The project is a 63.0 MW HEPP and it uses renewable sources to produce electricity. Since the installed capacity of the planned HEPP is larger than 15 MW; it is a large scale renewable energy project activity according to the Decision 17/ CP.7 Article 6. As per the justifications given above, the project activity is eligible under VCS.

The project is not a grouped project.

3.2 Baseline

- Approval of the baseline methodology:

The baseline for “Grid connected electricity generation from renewable sources: Uzundere I 63.0 MW Hydroelectric Power Plant Project, Turkey” project is established by using the UNFCCC official methodology ACM0002, version 11, namely “Consolidated baseline methodology for grid-connected electricity generation from renewable sources”.

For the calculation of the grid emission factor, UNFCCC Methodological Tool “Tool to Calculate Emission Factor of an Electricity System” version 02 is used.

For the assessment and demonstration of additionality, UNFCCC Methodological Tool “Tool for the Demonstration and Assessment of Additionality”, Version 05.2 is used.

Both of the tools are referred in the applied methodology.

The PD was submitted on 16 March 2010. Both the tools and the methodology are the latest available versions at the time of PD submission to the DOE, and are found appropriate by the validation team. Clean Development Mechanism Methodologies are approved under the VCS program.

- Correct application and justification of baseline methodology:

The choice of methodology ACM0002 version 11 is justified as the proposed project activity meets relevant applicability criteria:

- Uzundere I 63 MW Runoff-River HEPP Project is a grid connected renewable power generation project which adds electricity capacity from hydro power sources and which supplies electricity to a system that is supplied by at least one fossil fuel fired generating unit,
- The project activity results in a reservoir. But the power density of the power plant, as per definitions given in the Project Emissions section of ACM0002 version 11, is greater than 10 W/m² as calculated in section 1.3 “Estimated amount of emission reductions over the crediting period including project size” of the PD,

- The geographic and system boundaries for the relevant electricity grid can be clearly identified and information on the characteristics of the grid is available as proved in section 2.4 "Description of how the baseline scenario is identified and description of the identified baseline scenario" of the PD,
- The project does not involve switching from fossil fuels to renewable energy at the site of the project activity,
- The project involves construction of new units in a brand new plant, in other words the project does neither involve the addition of renewable energy generation units at an existing renewable power generation facility nor does it foresee to retrofit or modify an expired facility of renewable energy generation.

The baseline scenario was applied correctly and the justification for the choice of the baseline methodology is found appropriate by the validation team.

- Appropriate setting of baseline scenario

The baseline scenario is defined as follows:

"Electricity delivered to the grid by the project would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the Combined Margin (CM) calculations described in the "Tool to calculate emission factor for an electricity system"

The defined baseline scenario is in line with the methodology as the project activity is the installation of a new grid-connected renewable power plant.

As the project activity is not a retrofit or replacement of existing grid-connected renewable power plant, the step-wise procedure to identify the baseline given in ACM0002 version 11 is not used for this project activity.

The above setting of the baseline scenario is found appropriate by the validation team.

- Assessment and demonstration of additionality:

The assessment and demonstration of additionality of the project is done by using UNFCCC Methodological Tool "Tool for the Demonstration and Assessment of Additionality", Version 05.2. In applying this tool under Step 1a, the following four alternatives are identified as the alternative scenarios that are likely to happen in the absence of the project activity:

Alternative 1: Construction of a thermal (natural gas) power plant with the same annual power output

Alternative 2: The proposed project activity not undertaken as a VER project activity

Alternative 3: Construction of some other renewable energy plant with the same annual power output

Alternative 4: The same service of power supply is provided from the grid (Uzundere I HEPP is not built)

From the above mentioned alternatives 1, 2 and 4 are assessed to be realistic and credible alternatives. Alternative 3 is eliminated since the Black Sea Region is not very rich of wind and very steep terrain does not allow for wind power plants and the renewable energy source does not exists in the region for the geothermal energy.

Analyzing the alternatives that are defined as realistic, it is seen that all scenarios are consistent with the baseline definition of ACM0002 version 11. All of the

alternatives comply with legal requirements, as they are practices currently present in the host country.

The project proponent used Step 2 'Investment Analysis' for the demonstration of the additionality of the project activity.

Under Step 2-Investment Analysis there are four sub-steps.

Under sub-step 2a, Benchmark Analysis (Option III) is chosen and the other options are eliminated because, the proposed Project generates financial and economic benefits through the sales of electricity other than Voluntary Emissions Reduction (VCS-VER) related income. Therefore the simple cost analysis (Option I) cannot be taken. Investment comparison analysis (Option II) is only applicable to projects where alternatives should be similar investment projects. The alternative baseline scenario of the proposed project is the continuation of the current situation, where the equivalent electricity will be provided by the existing power units connected to the national grid. This method is also assessed to be appropriate by the validation team.

Under sub-step 2b, Equity IRR is selected as a financial indicator. The selected benchmark is assessed to be applicable to the type of IRR calculated as per the guidelines given in 'The tool for the demonstration and assessment of additionality'.

The country risk premium is taken from the study of Prof. Aswath Damodara. The Eurobond rate and risk premium value at the time of investment decision, which is the date of the Feasibility Study Report, is taken. This FSR dated May 2007 is the basis of the investment decision. The inputs of the IRR calculation are in accordance with the FSR. The benchmark calculation is given below:

Parameter		Chosen Value	Source
Rf	Risk-free rate	7.084%	The 30 year Eurobond which was issued by Treasury of Republic of Turkey was used as the proxy for the risk free rate in the country. This market return represents an after tax return over all components of the country's major index. The relevant interest at the date of investment decision of the facility which is assumed as the feasibility study of the facility completion date yield for this bond was 7.084% as of 30th of March 2007. (http://www.isyatirim.com.tr/report_download.aspx?file=4563)
β_i	The Beta Coefficient	0.922	The average of energy companies traded in ISE (Istanbul Stock Exchange) derived from one of the most prominent data provider Bloomberg.
$E(R_m) - R_f$	Market risk premium	9.29%	http://www.stern.nyu.edu/~adamodar/pc/archives/ctryprem07.xls
$E(R_i)$	Expected return on the capital asset	15.65%	-

Hence, the benchmark value is calculated as:

$$E(R_i) = R_f + (\beta_i \times E(R_m - R_f))$$

$$15.65\% = 7.084\% + (0.922 \times 9.29\%)$$

The benchmark is calculated using a globally accepted model and the data used for calculation is also applicable at the time of investment decision. Hence, the benchmark is assessed to be appropriate by the validation team and the financial expert.

Under sub-step 2c, the Equity IRR is calculated. Project proponents have also submitted an excel sheet showing the details of the calculations. Some revisions have been made to these calculations until the latest version, which was submitted to the validation team together with the final VCS PD.

The Feasibility Study Report dated May 2007 has been the basis of investment decision for the proposed project activity was submitted to the validation team. This was verified using a board meeting report dated 14.09.2007. The values used in the FSR are the unit values set by State Hydraulic Works and these values are applicable at the time of investment decision. The inputs of the IRR calculation are in accordance with the FSR.

The project proponents have calculated the IRR for 45 years from 2010 to 2045 until the license expiry date. This is on the conservative side as it reflects the reality and the calculations are not limited to the crediting period. The Project IRR is calculated as 8.16% that is lower than the benchmark, 15.65%.

The documents used to validate all the assumptions in IRR calculation are given below.

Parameters	Validated Value	Means of validation
Annual electricity generation	156,205 MWh	Feasibility Study Report
Electricity sales price	0.055 €/kWh	The minimum purchase guarantee of the State
Depreciation periods	Equipment : 30 years Construction: 40 years	Local rules and regulations
Parity of USD/TL	1.3777 \$/TL	Feasibility Study Report
Parity of USD/Euro	1.3618 \$/€	The average value is at the time of FSR. This value was validated through the Turkey Central Bank website.
Costs		
Regulator and Sedimentation Pool	3,742,844 \$	Feasibility Study Report
Conveyance Tunnels	31,664,734 \$	Feasibility Study Report
Approach Tunnels	1,829,118 \$	Feasibility Study Report
Siphon	62,076 \$	Feasibility Study Report
Loading Gate	514,233 \$	Feasibility Study Report
Penstock	11,482,967 \$	Feasibility Study Report
Power House Building	1,377,313 \$	Feasibility Study Report
Electromechanic Installments	18,900,000 \$	Feasibility Study Report
Hydromechanics Installments	1,053,744 \$	Feasibility Study Report
Transmission Lines	2,260,277 \$	Feasibility Study Report
Roads	6,169,703 \$	Feasibility Study Report
Construction Site and Permanent Structures	580,678 \$	Feasibility Study Report
Appropriation	399,216 \$	Feasibility Study Report
Etude, Consultancy and Project Costs	4,413,376 \$	Feasibility Study Report
General Production Expenses		
Operation and Maintenance Expense	1,404,181 \$	Feasibility Study Report
Renewal Expenses	300,443 \$	Feasibility Study Report

Under Sub-step 2d, sensitivity analysis is applied to the following items:

- Investment Costs
- Electricity Production
- Electricity Price

for a decrease of 5% and 10% and for an increase of 5% and 10%.

According to this sensitivity analysis;

When the investment costs decrease 10%, the IRR of the project activity rises up to 8.68% without VER revenues and up to 9.02% with 5 €/ton VER price. The IRR value decreases with the rise of the investment costs down to 7.69% without VER revenues and 8.09 with VER revenues at 5 €/ton VER price.

When electricity production raises 10%, the IRR of the project activity rises up to 9.63% without VER revenues and up to 10.01% with 5 €/ton VER price. The IRR value decreases with the decrease of the electricity price down to 6.72% without VER revenues and 6.97% with VER revenues at 5 €/ton VER price.

For the electricity price, variation on energy production was also applied in a bandwidth of +10% and -10% on the secondary energy production that reflects the volatility in calculations. Bandwidth of -10% and 0%, part of the production reflects the electricity production, which is 90%, guaranteed amount. Therefore, no sensitivity was applied to this part. When prices increase 10% the IRR goes up to 9.63% without VER and up to 9.98% with 5 €/ton VER price.

The above sensitivity analysis reveals the fact that no alternative scenario without VER revenues can make the project pass the benchmark IRR expectation that is 15.65%. Therefore, the project is not financially attractive.

The common practice analysis:

Under the common practice analysis section of the PD, hydro projects with similar installed capacity in the same region are compared to the proposed project activity.

First, this analysis is made for the Eastern Black Sea Region and on the basis of the installed capacity. It is seen with this analysis that the project activities that are present in the region are not yet operational and have very low completion rates (below 10%). When this analysis was extended to the host country, it was seen that the participation of private sector in electricity production from hydropower is a very new concept and only 57 projects out of 477 were completed more than 50% on the date of preparation of the PDD.

In addition, the new capacity from privately owned hydro facilities represents a friction of new facilities added to Turkey's capacity every year. On average, the new privately owned hydro facilities represent only 2.3% of new additions. Similarly, new additions from privately held hydro projects represents only 0.1% of Turkey's installed capacity in the last 6 years.

The validation team has concluded that the proposed project activity is not common practice in the region due to the reasons that are explained above.

3.3 Monitoring Plan

- *Approval of the monitoring methodology:*

The consolidated monitoring methodology for grid connected generation from renewable sources named as "Approved Monitoring Methodology ACM0002 version 11 is applied. All the applicability conditions of the monitoring methodology are the same as the applicability conditions of the baseline methodology that are given under section 3.2 of this validation report. The data will be archived electronically and be kept at least for 2 years after the

end of the last crediting period. All measurements will be conducted with calibrated measurement equipment according to relevant industry standards.

- Correct application and justification of the selected monitoring methodology:

The justification of the choice is explained in section 3.2 of this validation report. The monitoring methodology is applied correctly and appropriately.

- Information about monitoring plan is as follows:

The monitoring plan includes the monitoring of several parameters, the main parameter in the monitoring plan is the quantity of the net electricity that is delivered to the grid, and this parameter will be the basis for the emission reduction calculations.

According to the methodology ACM0002 version 11, the parameters used to calculate the Power Density of HEPPs also need to be monitored. These parameters are also included in the monitoring plan.

The first data that should be monitored is the amount of electricity ($EG_{\text{facility},y}$) that is annually fed into the grid by the plant. The project owner, KARADENİZ, has already signed an agreement with TEIAS that includes measuring the amount of electricity that goes to the system accurately. Based on this sales agreement, there is no need for an additional process or equipment to monitor the data.

The second data to be monitored is the installed capacity of the project activity. The data will be monitored from the supplier information on the equipments that shows the installed capacity of the turbines.

The third data to be monitored is the reservoir area. The reservoir area is expected to be topped when the water flow is peaked. The area can be calculated by making coordinate readings around the border of the reservoir. The coordinate readings will be done when the water flow is at its highest level

Also as the Combined Margin emission factor is calculated ex-ante, the parameters used in the calculation of the emission factor is also included in the monitoring plan, but the frequency of monitoring for these parameters are once for each crediting period as per the definitions in the "Tool to calculate the emission factor for an electricity system".

The below table shows the parameters that are included in the monitoring plan together with their monitoring process.

Data /Parameter	Description	Source of Data	Data Unit	Monitoring Frequency	QA/QC Procedures to be Applied
EG _{facility,y}	Quantity of net electricity delivered to the grid	Metering device from the grid operator	MWh	Continuous measurement and at least monthly recording	Double check with the grid operators
Cap _{pj}	Installed capacity of the project activity after implementation of the project activity	Project site	MW	Yearly	Supplier information on the related equipment and existence of the equipment will be checked
A _{pj}	Area of the reservoir measured in the surface of the water, after the implementation of the project activity, when the reservoir is full	Project site	m ²	Yearly	The readings will be done during the period when the water flow is high to have the largest reservoir area
EF _{Grid,CM,y}	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using the latest version of "Tool to calculate the emission factor for an electricity system"	As per the "Tool to calculate the emission factor for an electricity system"	tCO ₂ /MWh	Calculated once for each crediting period	-
FC _{i,y}	Amount of fossil fuel type I consumed in the project electricity system	As per the "Tool to calculate the emission factor for an electricity system"	Ton (m ³ for Natural Gas)	Calculated once for each crediting period	-
NCV _{i,y}	Net Calorific Value of fossil fuel types	As per the "Tool to calculate the emission factor for an electricity system"	TJ/Ton (TJ/m ³ for Natural Gas)	Calculated once for each crediting period	-

The data will be collected and stored by KARADENİZ not only during the crediting period but during also two years after the last issuance of VCUs to Uzundere I HEPP project for that crediting period.

The management of KARADENİZ is responsible with all monitoring and reporting issues in general.

Overall the monitoring methodology is found appropriate and suitable for this particular project activity.

3.4 Calculation of GHG Emissions

Anthropogenic emission reduction is a function of the expected net amount of electricity produced, for the given project, which amounts to 156,205 MWh per annum. The annual emission reduction is 87,318 tCO₂e. Over the period of ten years, the total amount of emission reduction will reach 873,180 tCO₂e. For all calculations, the rounded value of the combined emission factor, 0.559 tCO₂/MWh is applied.

Uzundere I HEPP consists of 2 unit turbines. Each unit has an output capacity of 31.6 MW. The project activity is implemented in 2 phases. On 27/05/2010, the power plant started operation with 1 turbine and on 07/10/2010, the remaining 1 turbine was commissioned.

Year	Electricity Output MWh	Baseline emission reductions in tones of CO ₂ e	Project emissions in tones of CO ₂ e	Annual estimation of emissions reductions in tones of CO ₂ e
2010	58,202	32,534	0	32,534
2011	156,205	87,318	0	87,318
2012	156,205	87,318	0	87,318
2013	156,205	87,318	0	87,318
2014	156,205	87,318	0	87,318
2015	156,205	87,318	0	87,318
2016	156,205	87,318	0	87,318
2017	156,205	87,318	0	87,318
2018	156,205	87,318	0	87,318
2019	156,205	87,318	0	87,318
2020	98,003	54,784	0	54,784
TOTAL				873,180

- *The appropriateness of the source, sink and reservoir:*

The Project emissions are calculated with the formula mentioned in ACM0002/Version11. The following calculation of the PD and the assumptions for the Project Emissions and Leakage made afterwards, is found acceptable and suitable for the project activity by the validation team.

The power density of the project activity is calculated as follows:

$$PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}$$

Where:

PD : Power density of the project activity, in W/m²

Cap_{PJ} : Installed capacity of the hydro power plant after the implementation of the project activity (W)

Cap_{BL} : Installed capacity of the hydro power plant before the implementation of the project activity (W). For new hydro power plants, this value is zero

A_{PJ} : Area of the reservoir measured in the surface of the water, after the implementation of the project activity, when the reservoir is full (m²)

A_{BL} : Area of the reservoir measured in the surface of the water, before the implementation of the project activity, when the reservoir is full (m^2). For new reservoirs, this value is zero

$$Cap_{PJ} = 63,000,000 \text{ W}$$

$$Cap_{BL} = 0$$

$$A_{PJ} = 7,275.54 \text{ m}^2$$

$$A_{BL} = 0$$

Therefore;

$$PD = (63,000,000 - 0) / (7,275.54 - 0) = 8,659.2 \text{ W/m}^2$$

Project emissions is assumed to be 0 since the power density of the project is greater than 10 W/m^2 .

No leakage emissions are considered. The main emissions potentially giving rise to leakage in the context of electric sector projects are emissions arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g. extraction, processing and transport). These emissions sources are neglected.

The baseline emissions BE_y are calculated as following:

$$BE_y = EG_y \times EF_{grid,CM,y}$$

Where:

BE_y = Baseline emissions in year y (tCO_2e/yr)

EG_y = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)

$EF_{grid,CM,y}$ = Baseline emission factor (tCO_2e/MWh)

Y = Refers to a given year

$$BE_y = 0.559 \times 156,205 = 87,318 \text{ tCO}_2e$$

Annual estimation of emissions reductions is calculated as follows in line with the ACM002/Version 11:

$$ER_y = BE_y - PE_y - LE_y$$

Where:

ER_y = Emission reductions in year y ($t \text{ CO}_2e/yr$)

BE_y = Baseline emissions in year y ($t \text{ CO}_2e /yr$)

PE_y = Project emissions in year y ($t \text{ CO}_2/yr$)

LE_y = Leakage emissions in year y ($t \text{ CO}_2/yr$)

Therefore, the annual emission reduction is estimated as:

$$ER_y = 87,318 - 0 - 0 = 87,318 \text{ tCO}_2e$$

- *The correctness and transparency of formulas and factors used,*

The formulas and factors used in the calculation of GHG emissions are found to be transparent and correct by the validation team. The emission factors of fuels that are used in the calculations are taken from the IPCC default values at the lower limit of uncertainty at a 95% confidence interval as provided in table 1.4 of Chapter 1 of Vol.2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories, as suggested by the methodology. The efficiency factors of power plants are taken from Annex 1 of the "Tool to calculate the emission factor for an electricity system". The efficiency factors used are on the conservative side.

- *The assumptions made for estimating GHG emission reductions*

The only assumption is the one about leakage emissions that is assumed to be zero as per the methodology. Other unknown parameters like the emission factors of fuels are taken from the IPCC values as mentioned above.

- *Uncertainties*

All data presented in the baseline calculations are provided from official sources. Therefore, uncertainties of data sets were not estimated. There are no uncertainties in the calculations.

3.5 Environmental Impact

The first feasibility study for the project which was designed for 43 MW of installed power by Karadeniz was originally finalized in June 2005. The project has "EIA not required certificate" dated 03/02/2006 that was given by the Ministry of Environment and Forestry. However, after this date the project capacity is increased and applied to Ministry of Environment and Forestry for review the certificate. The Ministry of Environment and Forestry has confirmed that after the capacity increased EIA is not required for this project. The "EIA not required certificate" dated 27/01/2009 was submitted to the validation team. Both of "EIA not required certificates" are stated under Annex 4 of the PD.

However, the certificates were cancelled by Rize Administrative Court and ordered the preparation of an EIA. An EIA Report was prepared by the project owner in 2010 and submitted to the Ministry of Environment and Forestry. On 01/04/2010 the Ministry of Environment and Forestry gave the EIA positive decision based on this EIA report. The "EIA positive certificate" is stated under Annex 5 of the PD.

The company made commitments to the Ministry of Environment and Forestry to minimize environmental impacts of the project. In addition to the above explanations, some of the commitments made by the company are as listed below:

- No waste will be thrown to the environment. The wastes will be disposed in line with the regulations.
- The excavated earth will be stored in line with the regulations and the construction site will be sprinkled to the decrease dust emissions.
- Sands and gravels will be provided from the region.
- There will not be any folding or grinding activities at construction site. Ready mixed concrete will be used in the construction of the project items. Therefore, there won't be dust emissions rising from these activities.
- There will be an infirmary at project site.

- The noise rising from the construction and operation of the plant will not exceed the allowed limits. The workers will work 8 hours a day and this will not have any impact on occupational health.

It has been verified depending on the processes that the project complies with the legal requirements.

3.6 Comments by stakeholders

The stakeholder meetings were required as part of the environmental assessments in compliance with EIA laws and regulations

In these meetings authorized people from KARADENİZ introduced project to the local people in terms of the size of the project, expected length of the construction period and to give details about how this project will impact their lives.

The project owner used different channels to invite comments by stakeholders. Most of the meetings were held through the Mukhtar (village governors) and Council of Elderly (Ihtiyar Heyeti; the village council) of Cataldere village, and the officials of Municipality of Kaptanpasa. The project owner visited various households in the project area to comprehend their needs and to identify how the project could assist in meeting the social and economic needs of the community.

The head of the village (mukhtar), Mustafa Karakas, addressed unemployment and weak economic activity as the major social problems in the area. The project owner was employed about 120 workers during the construction phase. Also, construction machines and equipment were hired from the vicinity of the project site. The local workers will be given priority during the hiring for plant operation.

The stakeholders asked for assistance for some social needs of Cataldere Village. The project owner helped local people and the institutions in many ways, from renewing the electricity infrastructure to building roads in the forest to help forest workers in fighting against possible fires. The village roads were maintained by the equipments and workers of the company. The company provided financial support for the mosque in Cataldere Village.

It was verified during the site visit through the interviews with local people and village heads that the project has a positive impact to the region's sustainable development. All of the stakeholders that were interviewed were content with the project activity and the general opinion was that the project was beneficial for the surrounding villages.

4 VALIDATION CONCLUSION

Bureau Veritas Certification has made the validation of "Uzuncayir 63.0 MW Hydroelectric Power Plant Project, in Rize, Turkey".

The validation was performed on the basis of VCS criteria and host country criteria and also on the criteria given to provide for consistent project operations, monitoring and reporting.

The validation consisted of the following three phases:

- i) Desk review of the project design and the baseline and monitoring plan; (March 2010)
- ii) Review of the baseline methodology by the specialist and the validator (March-December 2010)
- iii) Review of the investment analysis by the specialist and the validator (March-December 2010)
- iv) Follow-up interviews and on site visit; (12 April 2010)
- v) Resolution of outstanding issues and the issuance of the final validation report and opinion (December 2010)

Project participant/s used the latest tool for demonstration of the additionality. In line with this tool, the PD provides analysis of prevailing barriers to determine that the project activity itself is not the baseline scenario.

By generating electricity using renewable energy, the project is likely to result in reductions of GHG emissions partially displacing the electricity that would have been generated using fossil fuels. An analysis of the investment barrier and other barriers demonstrates that the proposed project activity is not a likely baseline scenario. Emission reductions attributable to the project are hence additional to any that would occur in the absence of the project activity. Given that the project is implemented and maintained as designed, the project is likely to achieve the estimated amount of emission reductions. The review of the Project Description (Version 01 dated 16 March 2010 until Version 07 dated 14 March 2011), the site visit and the subsequent follow-up interviews (please see the list of the persons interviewed under section 5.2 of this report) have provided Bureau Veritas Certification with sufficient evidence to determine the fulfillment of stated criteria. The Project Description was subsequently revised as Version 07 dated 14 March 2011 to resolve the issues that have risen during the interviews and subsequent interactions.

In our opinion, the project correctly applies and meets the relevant VCS requirements and the relevant host country criteria. The validation is based on the information made available to us and the engagement conditions detailed in this report.

5 REFERENCES

5.1. Documents

Documents provided by KARADENIZ that relates directly to the GHG components of the project and other reference documents are given below:

1. PD for "Grid connected electricity generation from renewable sources: Uzundere I 63.0 MW Hydroelectric Power Plant Project, Turkey"
 Version 01 dated 16 March 2010
 Version 02 dated 26 June 2010
 Version 03 dated 27 September 2010
 Version 04 dated 22 November 2010
 Version 05 dated 03 January 2011
 Version 06 dated 26 January 2011
 Version 07 dated 14 March 2011
2. Baseline Calculation Excel Sheet
 Version 01 dated 16 March 2010
 (Baseline_Karadeniz_Uzundere-I_16.03.2010)
 Version 02 dated 26 June 2010
 (Baseline_Karadeniz_Uzundere-I_26.06.2010)
 Version 03 dated 27 September 2010
 (Uzundere-I_Baseline_27 September 2010)
3. Investment Analysis – IRR Calculation Excel Sheet
 Version 01 dated 16 March 2010
 (Karadeniz Uzundere I _IRR)
 Version 02 dated 26 June 2010
 (Karadeniz Uzundere I _IRR v2.0 26 June 2010)
 Version 03 dated 27 September 2010
 (Karadeniz Uzundere I _IRR v3.0 27 September 2010)
 Version 04 dated 22 November 2010
 (Karadeniz Uzundere I _IRR v4.0 22 November 2010)
 Version 05 dated 26 January 2011
 (Karadeniz Uzundere I _IRR v5.0 22 November 2010)
 Version 06 dated 22 November 2010
 (Karadeniz Uzundere I _IRR v6.0 26 January 2011)
 Version 07 dated 26 January 2011
 (Karadeniz Uzundere I _IRR v7.0 14 March 2011)
4. Uzundere I HEPP Project Feasibility Report dated May 2007
5. EIA Report, 2010
6. Bank Letter, 14 October 2007
7. Court Decision numbered 2010/373
8. Provisional Acceptance Certificate approved by EMRA, 27 May 2010
9. Provisional Acceptance Certificate approved by EMRA, 07 September 2010

10. Board Decision, 14 September 2007
11. Approved consolidated baseline and monitoring methodology ACM0002 version 11
12. UNFCCC's Methodological Tool: "Tool for the demonstration and assessment of additionality", version 05.2
13. UNFCCC's Methodological Tool: "Tool to calculate the emission factor for an electricity system", version 02
14. VCS PD Template – 19 November 2007
15. VCS Validation Report Template – 19 November 2007
16. Voluntary Carbon Standard 2007.1

5.2. Persons Interviewed

List persons interviewed during the validation and site visit, or persons that contributed with other information that are not included in the documents listed above.

1. Mr. Kagan Patir, Carbon Consultant, GAIA Carbon Finance
2. Mr. Mustafa Karakas, Head of the Village
3. Mr. Huseyin Turan, Villager
4. Mr. Naci Ayar, Villager

ANNEX I – Validators' Competence

Lead Verifier: Ms. Bade Cebeci - Environmental Engineer

Bureau Veritas Certification Lead Verifier

Bade Cebeci has over 10 years experience in environmental sciences and auditing. She is a lead auditor for environment, safety and quality management systems. She is also lead verifier for GHG Emission Reduction Projects.

Verifier: Ms. Hande Sezer – Mining Engineer

Bureau Veritas Certification – Verifier

Hande Sezer has over 3 years experience in coal trade and 4 years of experience in various sectors. She is a verifier for GHG Emission Reduction Projects.

Verifier: Mrs. Isil Timuroglu – Environmental Engineer

Bureau Veritas Certification SAS– Verifier

Isil Timuroglu has over 5 years experience in environmental sectors. She worked about calculation of greenhouse gases. She is a verifier for GHG Emission Reduction Projects.

Baseline Specialist: Mrs. Yildiz Arikan - Assoc. Professor Dr

Sabancı University, Faculty of Management, Orhanlı, Tuzla, 34956, Istanbul, Turkey

Yildiz Arikan is an Electrical engineer and is working at Sabancı University. She has supported thesis related with energy . Also she has been conducting research studies on energy including "CO₂ Emission Research" Studies. Academically, Yildiz Arikan is working also on GHG project since 2005.

Investment Analysis Specialists:

Mr. Murat Gencer – Master of Economics

RiskTürk Software Development and Consultancy – Head of Financial Analysis Team

Murat Gencer, consultant and a trainer, has over 11 years of experience in FMCG, software development and banking sectors. He is specialized in project finance, financial modelling, risk management and MS Excel applications.

Internal Technical Reviewer: Mr. Ashok Mammen - PhD (Oils & Lubricants)

Bureau Veritas Certification - Internal Technical Reviewer

Over 20 years of experience in chemical and petrochemical field. Dr. Mammen is a lead auditor for environment, safety and quality management systems. He is also a lead verifier and tutor for GHG projects and has been involved in the validation and verification processes of more than 100 CDM/JI/VCS and other GHG projects.

ANNEX II - Validation Protocol

TABLE 1 VALIDATION REQUIREMENTS BASED ON THE VOLUNTARY CARBON STANDARD 2007.1

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
1. VCS Program specific issues					
a. Have the project(s) created another form of environmental credit (for example renewable energy certificates)?	VCS	3.1	The project has not created another form of environmental credit. This is stated under section 1.13 of the VCS PD.	OK	OK
b. If yes, have the project participants provided a letter from the program operator that the credit has not been used and has been cancelled from the relevant program?	VCS	3.1	N/A	OK	OK
c. Are the VCS PD, monitoring reports, and other documents required under the VCS Program in English?	VCS	3.2	All the documents that are presented in the PD are either in English or have English translations.	OK	OK
2. Project level requirements					
a. General requirements					
a. Have the project proponent applied an approved VCS Program methodology or a methodology from an approved GHG Program based on the list of current VCS Program approved GHG Programs and methodologies as set out on www.v-c-s.org ?	VCS	5.2	The project proponent has applied CDM approved Methodology ACM0002 version 10 and tools that are referred in ACM0002. However, ACM0002 version 10 is not the latest version of the methodology at the time of PD submission to the DOE. Please revise the PD using the latest version of the methodology.	CAR01	OK
b. Is the Project Start Date before 1 January 2002? (If yes, a CAR shall be raised as the Project Start Date for non-AFOLU projects for the VCS 2007.1 shall not be before 1 January	VCS	5.2.1	The project start date is not before 1 January 2002.	OK	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
2002)					
c. Will this validation be completed within two years of the Project Start Date? If not, was this validation contracted before 19 November 2008? (If yes validation shall be completed by 19 November 2009 and proof of contracting prior to 19 November 2008 shall be provided)	VCS	5.2. 1	This validation will be completed within two years of the project start date.	OK	OK
d. Is the earliest Project Crediting Period Start Date under the VCS 2007.1 28 March 2006 for non-AFOLU projects and 1 January 2002 for AFOLU projects ?	VCS	5.2. 1	The checklist question is N/A as the Project Crediting Period starts on July 1, 2010.	OK	OK
e. Does the project reduce GHG emissions from activities that are included in an emissions trading program; or take place in a jurisdiction or sector in which binding limits are established on GHG emissions?	VCS	5.2. 1	The project does not reduce GHG emissions from activities that are included in an emissions trading program; or take place in a jurisdiction or sector in which binding limits are established on GHG emissions. Turkey is not involved in any emission trading program other than the voluntary market and does not have any limits until 2012.	OK	OK
f. If yes, have the project proponents provided evidence that the reductions or removals generated by the project have or will not be used in the emissions trading program or for the purpose of demonstrating compliance with the binding limits that are in place in that jurisdiction or sector? [Such evidence could include: a letter from the program operator or designated national authority that emissions allowances (or other GHG credits used in the program) equivalent to the reductions or removals generated by the project have been	VCS	5.2. 1	N/A	OK	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
cancelled from the program; or national cap as applicable or; purchase and cancellation of GHG allowances equivalent to the GHG emissions reductions or removals generated by the project related to the program or national cap]					
g. Have project proponents claimed GHG credits from one project under more than one GHG Program? (If yes, a CAR shall be raised, as Project proponents shall not claim GHG credits from one project under more than one GHG Program)	VCS	5.2. 1	The project proponents have not claimed GHG credits from one project under more than one GHG Program. The project is not eligible for GS crediting as its capacity is over 20 MW. The project is not eligible for any of the compliance markets because of the position of the host country.	OK	OK
h. Was this project rejected by other GHG Programs?	VCS	5.2. 2	The project was not rejected by any other GHG program. Please refer to the above question.	OK	OK
i. If yes, have project proponents:	VCS	5.2. 2			
i. clearly stated in its VCS PD all GHG Programs for which the project has applied for credits and why the project was rejected? (Such information shall not be deemed commercially sensitive information	VCS	5.2. 2	N/A	OK	OK
ii. provided the VCS Program validator and verifier, VCS Program project database and VCS Program Registry with the actual rejection document(s) including any additional explanations?	VCS	5.2. 2	N/A	OK	OK
j. Is this a renewal of the Project Crediting Period?	VCS	5.2. 3	This validation is not a renewal of the first crediting period.	OK	OK
k. If yes have a VCS Program approved validator determined that the original project baseline	VCS	5.2.	N/A	OK	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
scenario(s) and additionality is still valid or has been updated taking account of new data and changed VCS Program requirements where applicable?		3			
<i>b. Standards and factors</i>					
a. Do standards and factors used to derive GHG emission data as well as any supporting data for additionality and baseline scenario(s) meet the following requirements:	VCS	5.5			
i. be publicly available from a reputable and recognised source (e.g. IPCC, published Government data etc)?	VCS	5.5	The standards and factors used to derive GHG emission data as well as any supporting data are taken from the IPCC and TEIAS that are both publicly available.	OK	OK
ii. be reviewed as part of its publication by a recognised competent organization?			TEIAS and IPCC are both reputable sources, and their publication processes are deemed reliable.	OK	OK
<i>c. Project grouping</i>					
a. Is this a grouped project?	VCS	5.6	This is not a grouped project, which is also stated under section 1.2 of the VCS PD. This was also verified during the site visit.	OK	OK
b. If yes , was this grouped project described in one VCS PD?	VCS	5.6	This is not a grouped project. The checklist question is N/A.	OK	OK
c. Does this PD include a description of the central GHG information system and controls associated with the project and its monitoring?	VCS	5.6	This is not a grouped project. The checklist question is N/A.	OK	OK
d. What is the sampling carried out by the VCS verifier?	VCS	5.6	As the project is not a grouped project no sampling was carried out. The checklist question is N/A.	OK	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
e. Have the sampling of a grouped project taken account of any sub groups and associated activities within each sub group?	VCS	5.6	As the project is not a grouped project no sampling was carried out. The checklist question is N/A.	OK	OK
f. Do this project, which intends to apply for the VCS Program VCUs as part of a grouped project also comply with the VCS Program requirements for grouped projects, detailed in the most recent version of the Program Guidelines 2007.1 on www.v-c-s.org ?	VCS	5.6	This is not a grouped project. The checklist question is N/A.	OK	OK
d. Content of the VCS PD					
a. Is the PD used as a basis for validation prepared in accordance with the latest template and guidance from the VCS?	VCS	5.7	The PD which is used as a basis for validation is prepared in accordance with the latest template and guidance from the VCS.	OK	OK
b. Is there a project title?	PD temp	1.1.	The project title is "Uzundere I 63.0 MW Hydroelectric Power Plant Project" as stated under section 1.1 of the PD. Document version: 01 Date of completion: 16.03.2010	OK	OK
c. Type/Category of the project	PD temp	1.2	The Project fits in: Sectoral Scope Number: 1 Sectoral Scope: Energy Industries – Renewable Energy	OK	OK
i. Is it defined whether the project category is part of a GHG program that has been approved by the VCS Board?	PD temp	1.2	Under section 1.2 of the VCS PD version 01, it is stated as follows: <i>"The project comes under Type I – Renewable Energy Project as per Appendix B of the procedures for CDM project activities."</i> CDM is approved by the VCS board.	OK	OK
ii. Is it specified if the project is a Grouped project?	PD temp	1.2	It is stated under section 1.2 of the VCS PD that the project is not a grouped project.	OK	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
d. Is the amount of emission reductions over the crediting period estimated, including project size? (Micro project: Less than 5,000 tonnes CO ₂ equivalent emissions reductions per year; Mega Project: More than 1,000,000 tonnes CO ₂ equivalent emissions reductions per year)	PD temp	1.3	Under section 1.3 of the VCS PD the amount of emission reductions over the crediting period is estimated including the project size. In VCS PD version 01, under section 1.3, it is estimated that the project will reduce 95,927.9 tons CO ₂ every year resulting in a total reduction of 959,279.0 tons of CO ₂ for the first crediting period. As the project will reduce more than 5,000 tons of CO ₂ and less than 1,000,000 tons of CO ₂ per year the project size Is defined as "Project" as per VCS rules.	OK	OK
e. Is a brief description of the project provided?	PD temp	1.4	The brief description of the project activity is provided under section 1.4 of the VCS PD. In this section the location and purpose of the project is given. The project's contribution to the region's sustainable development is given under section 1.16 of the VCS PD version 01. These contributions were verified during the site visit through the interviews with local people and village heads.	OK	OK
f. Is the project location, including geographic and physical information allowing the unique identification and delineation of the specific extent of the project, and including GPS project boundaries, provided?	PD temp	1.5	The detailed information on the project location, including geographic and physical information allowing the unique identification and delineation of the specific extent of the project are provided under section 1.5 of the VCS PD, however the following points needs further clarification: 1. Please provide the GPS coordinates separately for the power house and the	CL01	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
			reservoirs. 2. Please also identify the grid connection point for the power plant. 3. Please clarify what the second arrow represents in Figure 2. "Project Location on Turkey Map"		
g. Duration of the project activity/crediting period	PD temp	1.6			
i. Is the project start date, the date on which a financial commitment was made to the project and the project reached financial closure, provided?	PD temp	1.6	Under Section 1.6 of the VCS PD version 01, the following information is given; Ø The starting date of the project activity: July 1, 2010 when the plant commences electricity generation. Ø The date on which project reached financial closure: November 2006. Please also state the date on which a financial commitment was made to the project activity	CL02	OK
ii. Is the crediting period start date, i.e., the date the first monitoring period commenced, provided? (VCS project crediting period: A maximum of ten years which may be renewed at most two times)	PD temp	1.6	In VCS PD version 01, the crediting period start date is given as the same date of the starting date of the project activity, July 1, 2010; as per the requirements of VCS 2007.1.	OK	OK
h. Are the conditions prior to project initiation provided?	PD temp	1.7	Under section 1.7 of the VCS PD version 01, it is stated as follows: <i>"Prior to project initiation, the project area hosted some forestland and agricultural activities in small-scale lands."</i> The project is a Greenfield project which was	OK	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
			also verified during the site visit.		
i. Is a description of how the project will achieve GHG emission reductions and/or removal enhancements provided?	PD temp	1.8	Under section 1.8 of the VCS PD version 01, the description of how the project will achieve GHG emission reductions is given as follows: <i>"The existing electricity grid in Turkey is an interconnected single entity which to a larger extend fed by fossil fuel fired power plants. Uzundere I 63.0 MW HEPP Project is a grid connected renewable power generation project which adds electricity capacity from hydro power sources which otherwise would have been generated by the fossil fuel fired power plant."</i>	OK	OK
j. Are project technologies, products, services and the expected level of activity described?	PD temp	1.9	The project technologies are given under section 1.9 of the VCS PD. Also information on the products, services and expected level of activity is given under the same section. Under this section it is stated that the turbines will be supplied from Alstom, and it is also stated that the expected lifetime of the equipment is around 20 years. Please clarify the above mentioned statements as the lifetime of the Alstom turbines are usually longer than 20 years.	CL03	OK
k. Does the VCS PD include identification of relevant local laws and regulations related to the project and demonstration of compliance with them?	PD temp	1.10	The applicable laws and regulations are listed under sections 1.10 and 2.5. It is stated under section 1.10 that the project activity is in compliance of these laws and regulations. This statement was verified during the site visit by seeing the related permits.	OK	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
l. Are risks that may substantially affect the project's GHG emission reductions or removal enhancements identified?	PD temp	1.11	In VCS PD version 01, the following risks are identified: § decrease of rainfall in the region due to the climate change § Technical risks: The risks that stem from misoptimization of the installed power from water flow as well as choosing the not best-fit turbines and technical equipment for the design. Please clarify how the identified technical risk has a potential to effect projects' GHG emission reductions. The installed power of the project activity has already been changed from 43.68 MW to 63.0 MW and the project activity uses Alstom Turbines, which are optimized taking into consideration the flow regime.	CL04	OK
m. Is confirmation that the project was not implemented to create GHG emissions primarily for the purpose of its subsequent removal or destruction demonstrated?	PD temp	1.12	Under section 1.12 of the PD version 01 it is stated that <i>"the project is not implemented to create GHG emissions primarily for the purpose of its subsequent removal or destruction"</i>	OK	OK
n. Has the project created another form of environmental credit (for example renewable energy certificates)?	PD temp	1.13	The project has not created another form of environmental credit. This is stated under section 1.13 of the VCS PD.	OK	OK
o. If yes, has the proponent provided a letter from the program operator that the credit has not been used and has been cancelled from the relevant program?	PD temp	1.13	N/A	OK	OK
p. Was the project rejected under other GHG programs (if applicable)	PD temp	1.14	The project has not been rejected under any other GHG programs; this was also stated under section 1.14 of the VCS PD.	OK	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
			The project is not eligible for GS crediting, as it is a large scale HEPP. The project is not eligible for any of the compliance markets because of the position of the host country.		
q. If yes, does the project:	PD temp	1.14			
i. clearly state in its VCS PD all GHG programs for which the project has applied for credits and why the project was rejected? (Such information shall not be deemed commercially sensitive information)	PD temp	1.14	N/A	OK	OK
ii. provide the VCS verifier and Registry with the actual rejection document(s) including explanation?	PD temp	1.14	N/A	OK	OK
r. Are project proponents roles and responsibilities, including contact information of the project proponent, other project participants provided?	PD temp	1.15	The contact information of the project proponent and carbon consultants are given in a tabular format under section 1.15 of the VCS PD. Please also briefly define the roles and responsibilities of the contact persons regarding the carbon crediting process.	CL05	OK
s. Is any information relevant for the eligibility of the project and quantification of emission reductions or removal enhancements, including legislative, technical, economic, sectoral, social, environmental, geographic, site-specific and temporal information provided?	PD temp	1.16	Information relevant for the eligibility of the project and quantification of emission reductions or removal enhancements, including legislative, technical, economic, sectoral, social, environmental, geographic, site-specific and temporal information is provided under section 1.16 of the VCS PD version 01. These contributions were verified during the site visit.	OK	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
t. Is there any commercially sensitive information that has been excluded from the public version of the VCS PD that will be displayed on the VCS Project Database?	PD temp	1.17	Section 1.17 "List of commercially sensitive information" is missing in the PD version 01.	CAR02	OK
u. If yes, was it listed?	PD temp	1.17	N/A	OK	OK
v. Are title and reference of the VCS methodology applied to the project activity and explanation of methodology choices provided?	PD temp	2.1	Under section 2.1 of the VCS PD it is stated that the project proponent has applied CDM approved Methodology ACM0002 version 10 and tools that are referred in ACM0002. However, ACM0002 version 10 is not the latest version of the methodology at the time of PD submission to the DOE. Please revise the PD using the latest version of the methodology.	CAR01	OK
w. Does the project use one of the VCS program approved project methodologies and provide information relevant to methodology deviations or methodology revisions?	PD temp	2.1	The project uses a Clean Development Mechanism approved methodology without any deviations or revisions to the existing methodology. UNFCCC Clean Development Mechanism methodologies are approved under the VCS.	OK	OK
x. Are the choice of the methodology and its applicability to the project activity justified?	PD temp	2.2	Justification of the choice of the methodology is stated in section 2.2 of the VCS PD version 01. The project is an installation a new hydropower plant where no renewable power plant was operated prior to the implementation of the project activity (greenfield plant). The project activity results in construction of a new reservoir with a power density greater than 4 W/m ² .	OK	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
y. Are GHG sources, sinks and reservoirs identified for the baseline scenario and for the project?	PD temp	2.3	The project activity does not use fossil fuel Under section 2.4 of the PD version 01, the GHG sources are identified for both the baseline scenario and the project activity. However in the table CH₄ emissions from the reservoir has been considered twice. For the operation and construction of the HEPP CH₄ emissions are considered to be negligible, but it is stated on the next row that CH₄ emissions from reservoir is a main emission source. Also under this section the same information with the exact same sentences are given twice. Please clarify.	CL06	OK
z. Is it described how the baseline scenario is identified and the identified baseline scenario?	PD temp	2.4	Under section 2.4 of the VCS PD version 01 it is stated that the baseline scenario is identified according to the methodology ACM0002 which is as follows: <i>"Since the project is an installation of a new grid-connected renewable power plant, the baseline scenario is formulated in ACM0002 / Version 10: "Electricity delivered to the grid by the project would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the Combined Margin (CM) calculations"</i> It is also approved by the validation team that the selected baseline scenario is in line with the methodology. (both version 10 and 11)	OK	OK
aa. Has the project proponent selected the most	PD	2.4	Yes. The project proponent has selected the	OK	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
reasonable baseline scenario for the project?	temp		most reasonable baseline scenario for the project activity.		
bb. Does it reflect what most likely would have occurred in the absence of the project?	PD temp	2.4	The baseline scenario reflects what most likely would have occurred in the absence of the project activity.	OK	OK
cc. Is it described how the emissions of GHG by source in baseline scenario are reduced below those that would have occurred in the absence of the project activity (assessment and demonstration of additionality)?	PD temp	2.5	Under section 2.5 of the VCS PD version 01, it is described how the GHG emissions by source are reduced below those that would have occurred in the absence of the project activity using the latest version (version 05.2) of the UNFCCC Methodological Tool "Tool for the demonstration and assessment of additionality" as suggested by the methodology ACM0002. For further information please refer to "Table 3: Checklist based on Methodological tool for assessment of additionality (version 05.2 - EB 39 report Annex 10)"	OK	OK
dd. Has the project proponent demonstrated, in the VCS PD, in addition to describing how the project meets the VCS methodology, that the project is additional based on one of the tests, the project test, the performance test, and technology test?	PD temp	2.5	The project proponents have used the UNFCCC Approved consolidated baseline and monitoring methodology ACM0002 "Consolidated baseline methodology for grid-connected electricity generation from renewable sources" and the additionality tool referred in this methodology. Therefore the checklist question is N/A.	OK	OK
ee. Are title and reference of the VCS methodology (which includes the monitoring requirements) applied to the project activity and explanation of methodology choices provided?	PD temp	3.1	Under section 3.1 of the VCS PD the title and reference of the VCS methodology applied to the project activity is given as follows: <i>"The consolidated monitoring methodology for grid connected generation from renewable sources named as "Approved Monitoring</i>	CAR01	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
			<i>Methodology ACM0002 / Version 10 is applied. The monitoring methodology is applicable as the data will be archived electronically and be kept at least for 2 years after the end of the last crediting period.</i> ACM0002 version 10 is not the latest version of the methodology at the time of PD submission to the DOE. Please revise the PD using the latest version of the methodology.		
ff. Is monitoring, including estimation, modelling, measurement or calculation approaches describe including:	PD temp	3.2			
i. Purpose of monitoring?	PD temp	3.2	In the PD version 01, under section 3.2 the purpose of the monitoring is given as follows: <i>"This monitoring plan is prepared as a road map for the verification of the information which needs to be monitored periodically"</i>	OK	OK
ii. Types of data and information to be reported, including units of measurement?	PD temp	3.2	Types of data and information to be reported including the units of measurement is given under section 3.2 of the VCS PD version 01 in a tabular format; however the parameter $FC_{i,y}$ is defined as: <i>"Turkey's gross electricity generated by primary energy source of the five most recent years"</i> Please clarify.	CL07	OK
iii. Origin of the data?	PD temp	3.2	The origin of the data is given under section 3.2 of the VCS PD version 01 in a tabular format. The source of data is also given under section	CAR03	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
			3.3 of the VCS PD version 01 within the data/parameter tables. For the parameter $EG_{\text{facility},y}$ the source of data is given as "metering device from the grid operator" under section 3.2 and as "monthly invoice" under section 3.3. In the methodology the source of data for this parameter is defined as "Project activity site" Please define the same source for the same parameter and make sure the source is in line with the methodology.		
iv. Monitoring, including estimation, modelling, measurement or calculation approaches?	PD temp	3.2	Information about the monitoring is given under section 3.2 of the VCS PD version 01 in a tabular format.	OK	OK
v. Monitoring times and periods, considering the needs of intended users?	PD temp	3.2	Monitoring period is given under section 3.2 of the VCS PD version 01 in a tabular format.	OK	OK
vi. Monitoring roles and responsibilities ?	PD temp	3.2	Monitoring roles and responsibilities have been defined in the monitoring plan under section 3.2 of the VCS PD version 01.	OK	OK
vii. Managing data quality?	PD temp	3.2	The data will be archived electronically and also in paper form. QA/QC procedures for each data/parameter have also been defined under section 3.1 of the VCS PD version 01.	OK	OK
gg. Are data and parameters monitored/selecting relevant GHG sources, sinks and reservoirs for monitoring or estimating GHG emissions and removals described in the tabular form including:	PD temp	3.3			
i. Data unit?	PD	3.3	In the VCS PD version 01, data units are given	OK	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
	temp		for the parameters that are included in the monitoring plan as per the methodology ACM0002.		
ii. Description?	PD temp	3.3	In the VCS PD version 01, the descriptions for the parameters are given as per the methodology ACM0002.	OK	OK
iii. Source of data to be used?	PD temp	3.3	The origin of the data is given under section 3.2 of the VCS PD version 01 in a tabular format. The source of data is also given under section 3.3 of the VCS PD version 01 within the data/parameter tables. For the parameter $EG_{\text{facility},y}$ the source of data is given as "metering device from the grid operator" under section 3.2 and as "monthly invoice" under section 3.3. In the methodology the source of data for this parameter is defined as "Project activity site" Please define the same source for the same parameter and make sure the source is in line with the methodology.	CAR03	OK
iv. Value of data applied for the purpose of calculating expected emissions reductions?	PD temp	3.3	In the VCS PD version 01 the value applied for the parameter CAP_{PJ} is given as 20.5 MW, however it is stated in other sections of the PD that the installed capacity of the project activity is 63 MW. Please clarify. Also for the parameter A_{PJ} the value is given as 13,837 m². This parameter was given as 13.4 km² in some other section of the PD. Please clarify the Reservoir Area and please also	CL08	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
			define where this value is taken from. Please include all of your references in the PD.		
v. Description of measurement methods and procedures to be applied?	PD temp	3.3	Measurement procedures for each data/parameter is given in the tabular format, however for the parameter $EG_{\text{facility},y}$ the measurement procedure is given as MWh. Please define the measurement procedures as per the methodology.	CL09	OK
vi. QA/QC procedures to be applied?	PD temp	3.3	QA/QC procedures to be applied for the parameters that are included in the monitoring plan are given in VCS PD version 01 as per the methodology.	OK	OK
vii. Any comment?	PD temp	3.3	No comment is given for the parameters included in the monitoring plan in VCS PD version 01.	OK	OK
hh. Is the monitoring plan described?	PD temp	3.4	The monitoring plan is described in detail under section 3.4 of the VCS PD version 01.	OK	OK
ii. Are methodological choices explained?	PD temp	4.1	Methodological choices are explained under section 4.1 of the VCS PD version 01. ACM0002 version 10 is not the latest version of the methodology at the time of PD submission to the DOE. Please revise the PD using the latest version of the methodology. Please also include the version of the "Tool to calculate the emission factor for an electricity system"	CAR01 CL10	OK OK
jj. Are GHG emissions and/or removals for the baseline scenario quantified?	PD temp	4.2	GHG emissions for the baseline scenario are quantified under section 2.4 of the VCS PD	OK	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
			version 01.		
kk. Are GHG emissions and/or removals for the project quantified?	PD temp	4.3	The GHG emissions for the project (the Power Density calculations) are given under section 2.4 of the VCS PD version 01.	OK	OK
ll. Are GHG emission reductions and removal enhancements for the GHG project quantified?	PD temp	4.4	The GHG emission reductions for the GHG project are quantified under section 2.4 of the PD version 01.	OK	OK
mm. Was a summary of environmental impact assessment, when such an assessment is required by applicable legislation or regulation, provided? ISO14064-2, 5.2.k	PD temp	5	A summary of the environmental impacts is given under section 5 of the VCS PD version 01. However in this section it is stated that the domestic law and regulation does not require any environmental impact assessment. Please clarify as this is not the case at the moment. Please also include a summary of the EIA prepared by the project owner which was also sent to the validation team.	CL11	OK
nn. Were relevant outcomes from stakeholder consultations and mechanisms for on-going communication provided?	PD temp	6	Relevant outcomes from stakeholder consultations are provided under section 6 of the VCS PD version 01. The stakeholders comments were also taken during the site visit via interviews. All the stakeholders that were interviewed were content about the project activity.	OK	OK
oo. Was a chronological plan for the date of initiating project activities, date of terminating the project, frequency of monitoring and reporting and the project period, including relevant project activities in each step of the GHG project cycle provided?	PD temp	7	A chronological plan is given under section 7 of the VCS PD version 01.	OK	OK
pp. Was evidence of proof of title provided through	PD	8.1			

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
one of the following:	temp				
i. a legislative right?	PD temp	8.1	The generation license was given under Annex 2 as a proof of title.	OK	OK
ii. a right under local common law?	PD temp	8.1	N/A	OK	OK
iii. ownership of the plant, equipment and/or process generating the reductions/removals?	PD temp	8.1	N/A	OK	OK
iv. a contractual arrangement with the owner of the plant, equipment or process that grants all reductions/removals to the proponent?	PD temp	8.1	N/A	OK	OK
qq. Does the project reduce GHG emissions from activities that participate in an emissions trading program, or take place in a jurisdiction or sector in which binding limits are established on GHG emissions?	PD temp	8.2	N/A	OK	OK
rr. If yes, have project proponents provided evidence that the reductions or removals generated by the project have or will not be used in the Program or jurisdiction for the purpose of demonstrating compliance, such as:	PD temp	8.2	N/A	OK	OK
i. a letter from the Program operator or designated national authority that emissions allowances (or other GHG credits used in the Program) equivalent to the reductions/removals generated by the project have been cancelled from the Program; or national cap as applicable?	PD temp	8.2	N/A	OK	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
ii. purchase and cancellation of GHG allowances equivalent to the reductions/removals generated by the project related to the Program or national cap?	PD temp	8.2	N/A	OK	OK
e. Additionality					
a. Has the project proponent demonstrated that the project is additional using one of the following tests: Test 1 - The project test; Test 2 - Performance test; Test 3 - Technology test?	VCS	5.8	The project proponents have used the UNFCCC Approved consolidated baseline and monitoring methodology ACM0002 "Consolidated baseline methodology for grid-connected electricity generation from renewable sources" and the additionality tool referred in this methodology. According to Frequently Asked Questions sections on the VCS website, the VCS additionality guidance is not a substitute for the additionality requirements set out in the methodology being used. Therefore the checklist question is N/A. The additionality of the project activity is assessed using "Table 2: Validation requirements based on the Validation and Verification Manual V 01.1 (EB51 Annex 3)" of this protocol.	OK	OK
b. If the project proponent used Test 1:	VCS	5.8			
iii. Step 1 - Regulatory Surplus - Is the project be mandated by any enforced law, statute or other regulatory framework? (If yes a CAR shall be issued and the project shall be deemed non additional).	VCS	5.8	Please refer to question a above. The checklist question is N/A. The additionality of the project activity is assessed using Table 2 of this protocol.	OK	OK
iv. Step 2 - Implementation Barriers - Does the project face one (or more) distinct	VCS	5.8	Please refer to question a above. The checklist question is N/A. The additionality of the project	OK	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
barrier(s) compared with barriers faced by alternative projects?			activity is assessed using Table 2 of this protocol.		
i. Investment Barrier – Does the project face capital or investment return constraints that can be overcome by the additional revenues associated with the generation of VCUs?	VCS	5.8	Please refer to question a above. The checklist question is N/A. The additionality of the project activity is assessed using Table 2 of this protocol.	OK	OK
ii. Technological Barriers – Does the project face technology-related barriers to its implementation?	VCS	5.8	Please refer to question a above. The checklist question is N/A. The additionality of the project activity is assessed using Table 2 of this protocol.	OK	OK
iii. Institutional barriers – Does the project face financial, organizational, cultural or social barriers that the VCU revenue stream can help overcome?	VCS	5.8	Please refer to question a above. The checklist question is N/A. The additionality of the project activity is assessed using Table 2 of this protocol.	OK	OK
v. Step 3 – Common Practice	VCS	5.8			
i. Is project type common practice in sector/region, compared with projects that have received no carbon finance?	VCS	5.8	Please refer to question a above. The checklist question is N/A. The additionality of the project activity is assessed using Table 2 of this protocol.	OK	OK
ii. if it is common practice, have the project proponents identified barriers faced compared with existing projects?	VCS	5.8	Please refer to question a above. The checklist question is N/A. The additionality of the project activity is assessed using Table 2 of this protocol.	OK	OK
iii. Is the demonstration that the project is not common practice based on guidance in the GHG Protocol for Project Accounting,	VCS	5.8	Please refer to question a above. The checklist question is N/A. The additionality of the project activity is assessed using Table 2 of this	OK	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
Chapter 7?			protocol.		
iv. Was it collected data on all baseline candidates within the geographic area and calculating a relative percentage for each different technology or practice? Common practice refers to the predominant technologies or practices in a given market, as determined by the degree to which those technologies or practices have penetrated the market (defined by a specified geographic area). This percentage could be based on the number of plants or sites using each technology or practice, or could be Weighted by the proportion of the total output for the market that is attributed to each technology or practice.	GHG PROT OCOL	7.4. 2 AND 7.6	Please refer to question a above. The checklist question is N/A. The additionality of the project activity is assessed using Table 2 of this protocol.	OK	OK
c. If the project proponent used Test 2:	VCS	5.8			
vi. Step 1 – Regulatory Surplus - Is the project be mandated by any enforced law, statute or other regulatory framework? (If yes a CAR shall be issued and the project shall be deemed non additional).	VCS	5.8	Please refer to question a above. The checklist question is N/A. The additionality of the project activity is assessed using Table 2 of this protocol.	OK	OK
vii. Step 2: Performance Standard	VCS	5.8			
i. Are the emissions generated per unit output by the project below the level that has been approved by the VCS Program for the product, service, sector or industry, as the level defined to	VCS	5.8	Please refer to question a above. The checklist question is N/A. The additionality of the project activity is assessed using Table 2 of this protocol.	OK	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
ensure that the project is not business-as-usual?					
ii. Are performance standard based additionality tests approved through the double approval process and by the VCS Board? (The list of approved performance standards is on www.v-c-s.org)	VCS	5.8	Please refer to question a above. The checklist question is N/A. The additionality of the project activity is assessed using Table 2 of this protocol.	OK	OK
d. If the project proponent used Test 3:	VCS	5.8			
viii. Step 1: Regulatory Surplus - Is the project be mandated by any enforced law, statute or other regulatory framework? (If yes a CAR shall be issued and the project shall be deemed non additional).	VCS	5.8	Please refer to question a above. The checklist question is N/A. The additionality of the project activity is assessed using Table 2 of this protocol.	OK	OK
ix. Step 2: Technology Additionality – Are the project and its location contained in the list of project types and applicable areas approved as being additional by the VCS Program? (The approved list is available on www.v-c-s.org)	VCS	5.8	Please refer to question a above. The checklist question is N/A. The additionality of the project activity is assessed using Table 2 of this protocol.	OK	OK
f. Identifying GHG sources, sinks and reservoirs relevant to the project					
a. Refer to Clause 6, under Methodologies.	VCS	5.9	Refer to Clause 6, under Methodologies.	OK	OK
g. Determining the baseline scenario					
a. Has the project proponent selected the most conservative baseline scenario for the project, based on the requirements in the applicable VCS methodology?	VCS	5.10	The project proponent has used ACM0002 Methodology. According to ACM0002 for greenfield renewable power plants the only applicable baseline	OK	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
			scenario is as follows: Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "Tool to calculate the emission factor for an electricity system" Under section 2.4 of the VCS PD version 01, the project proponent has also defined the above mentioned baseline scenario for the project activity.		
b. Does the baseline scenario set out the geographic scope as applicable to the project?	VCS	5.10	The baseline scenario is determined according to the methodology ACM0002 therefore the checklist question is N/A. The appropriateness of the baseline scenario and the calculations are assessed using "Table 2: Validation requirements based on the Validation and Verification Manual V 01.1 (EB51 Annex 3)" of this protocol.	OK	OK
c. Has the project participant demonstrated that the project has met the requirements in ISO 14064-2:006 clause 5.4?	VCS	5.10	The baseline scenario is determined according to the methodology ACM0002 therefore the checklist question is N/A. The appropriateness of the baseline scenario and the calculations are assessed using "Table 2: Validation requirements based on the Validation and Verification Manual V 01.1 (EB51 Annex 3)" of this protocol.	OK	OK
i. Does the project proponent select or	ISO	5.4	The baseline scenario is determined according	OK	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
establish criteria and procedures for identifying and assessing potential baseline scenarios considering the following ? a) the project description, including identified GHG sources, sinks and reservoirs (see 5.3); b) existing and alternative project types, activities and technologies providing equivalent type and level of activity of products or services to the project; c) data availability, reliability and limitations; d) other relevant information concerning present or future conditions, such as legislative, technical, economic, sociocultural, environmental, geographic, site-specific and temporal assumptions or projections.	1406 4.2		to the methodology ACM0002 therefore the checklist question is N/A. The appropriateness of the baseline scenario and the calculations are assessed using "Table 2: Validation requirements based on the Validation and Verification Manual V 01.1 (EB51 Annex 3)" of this protocol.		
ii. Does the project proponent demonstrate equivalence in type and level of activity of products or services provided between the project and the baseline scenario and shall explain, as appropriate, any significant differences between the project and the baseline scenario ?	ISO 1406 4.2	5.4	The baseline scenario is determined according to the methodology ACM0002 therefore the checklist question is N/A. The appropriateness of the baseline scenario and the calculations are assessed using "Table 2: Validation requirements based on the Validation and Verification Manual V 01.1 (EB51 Annex 3)" of this protocol.	OK	OK
iii. Does the project proponent select or establish, explain and apply criteria and procedures for identifying and justifying the baseline scenario ?	ISO 1406 4.2	5.4	The baseline scenario is determined according to the methodology ACM0002 therefore the checklist question is N/A. The appropriateness of the baseline scenario and the calculations are assessed using "Table 2: Validation requirements based on the Validation and Verification Manual V 01.1 (EB51 Annex 3)" of this protocol.	OK	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
iv. In developing the baseline scenario, does the project proponent select the assumptions, values and procedures that help ensure that GHG emissions reductions or removal enhancements are not over-estimated ?	ISO 1406 4.2	5.4	The baseline scenario is determined according to the methodology ACM0002 therefore the checklist question is N/A. The appropriateness of the baseline scenario and the calculations are assessed using "Table 2: Validation requirements based on the Validation and Verification Manual V 01.1 (EB51 Annex 3)" of this protocol.	OK	OK
v. Does the project proponent select or establish, justify and apply criteria and procedures for demonstrating that the project results in GHG emissions reductions or removal enhancements that are additional to what would occur in the baseline scenario ?	ISO 1406 4.2	5.4	The baseline scenario is determined according to the methodology ACM0002 therefore the checklist question is N/A. The appropriateness of the baseline scenario and the calculations are assessed using "Table 2: Validation requirements based on the Validation and Verification Manual V 01.1 (EB51 Annex 3)" of this protocol.	OK	OK
d. Has the project participant demonstrated that the project has met all relevant regulations, legislation and project approvals (e.g. environmental permits)?	VCS	5.10	The project participant has demonstrated that the project has met all relevant regulations. The relevant permits have been reviewed during the site visit, and also copies of some permits have been submitted to the validation team as a proof of compliance with the relevant laws and regulations.	OK	OK
<i>h. Monitoring the GHG project</i>					
a. Has the project proponent established and maintained criteria and procedures for obtaining, recording, compiling and analysing data and information important for quantifying and reporting GHG emissions and/or removals relevant for the project and baseline scenario	VCS	5.11	The project proponent has established criteria and procedures for obtaining, recording, compiling and analysing data and information important for quantifying and reporting GHG emission reductions relevant for the project and baseline scenario.	OK	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
(i.e. GHG information system)?			Whether the project proponent maintains these procedures shall be checked during the verification of the project activity.		
b. Do the monitoring procedures include?	VCS	5.11			
i. purpose of monitoring?	VCS	5.11	Please refer to question ff-i of this checklist. The application of the monitoring procedures will be checked during the verification of the project activity.	OK	OK
ii. types of data and information to be reported, including units of measurement?	VCS	5.11	Please refer to question ff-ii of this checklist. The application of the monitoring procedures will be checked during the verification of the project activity.	OK	OK
iii. origin of the data?	VCS	5.11	Please refer to question ff-iii of this checklist. The application of the monitoring procedures will be checked during the verification of the project activity.	OK	OK
iv. monitoring methodologies, including estimation, modelling, measurement or calculation approaches?	VCS	5.11	Please refer to question ff-iv of this checklist. The application of the monitoring procedures will be checked during the verification of the project activity.	OK	OK
v. monitoring times and periods, considering the needs of intended users?	VCS	5.11	Please refer to question ff-v of this checklist. The application of the monitoring procedures will be checked during the verification of the project activity.	OK	OK
vi. monitoring roles and responsibilities?	VCS	5.11	Please refer to question ff-vi of this checklist. The application of the monitoring procedures will be checked during the verification of the project activity.	OK	OK
vii. GHG information management systems,	VCS	5.11	Please refer to question ff-vii of this checklist.	OK	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
including the location and retention of stored data?			The application of the monitoring procedures will be checked during the verification of the project activity.		
c. Where measurement and monitoring equipment is used, does the project proponent ensure the equipment is calibrated according to current good practice?	VCS	5.11	It is stated under section 3.1 of the PD version 01 that: <i>"All measurements will be conducted with calibrated measurement equipment according to relevant industry standards."</i> The calibration records of all the equipment used in monitoring shall be checked during the verification of the project activity.	OK	OK
d. Does the project proponent apply GHG monitoring criteria and procedures on a regular basis during project implementation?	VCS	5.11	The project is at validation stage therefore the checklist question is N/A.	OK	OK
<i>i. Monitoring reports for the GHG project</i>					
a. Do monitoring reports include all the monitoring data, calculations, estimations, conversion factors and others standard factors as defined in the monitoring clause of the applied VCS Program methodology and set out in the VCS PD? (A list of VCS approved methodologies is available on www.v-c-s.org)	VCS	5.12	The project is at validation stage therefore the checklist question is N/A.	OK	OK
<i>j. Records relating to the project</i>					
a. Has the project proponent kept all documents and records in a secure and retrievable manner for at least two years after the end of the project crediting period.?	VCS	5.13	The project is at validation stage therefore the checklist question is N/A.	OK	OK
<i>k. Information to validator and verifier</i>					
a. Has the project proponent made available to	VCS	5.14	Yes, the project proponent has made the VCS	OK	OK

CHECKLIST QUESTION	Ref.	item	COMMENTS	Draft Concl	Final Concl
the validator the VCS PD, proof of title and any requested supporting information and data needed to evidence statements and data in the VCS PD and proof of title?			PD, proof of title and all the information and data available to the validation team.		
b. Has the project proponent made the VCS PD and validation report available to the verifiers as well as a monitoring report applicable to the period of monitoring and any requested supporting information and data needed to evidence statements and data in the monitoring report?	VCS	5.14	The project is at validation stage therefore the checklist question is N/A.	OK	OK

TABLE 2 VALIDATION REQUIREMENTS BASED ON THE VALIDATION AND VERIFICATION MANUAL V 01.1 (EB51 ANNEX 3)

CHECKLIST QUESTION	Ref.	§	COMMENTS		Draft Concl	Final Concl
			COUNTRY A	COUNTRY B		
1. Approval						
1.1. Has the DNA of each Party indicated as being involved in the proposed CDM project activity in section A.3 of the PDD provided a written letter of approval? State the country.	VVM	45	The project is not developed under compliance markets. Therefore the checklist question is N/A.		OK	OK
1.2. Does the letter of approval from DNA of each Party confirm that : - The Party is a Party of the Kyoto Protocol - The participation is voluntary - In the case of the host Party, the proposed CDM project activity contributes to the sustainable development of the country - Refers to the precise proposed CDM project activity title in the PDD being submitted for registration	VVM	45	The project is not developed under compliance markets. Therefore the checklist question is N/A.		OK	OK
1.3. Is(are) the letter(s) of approval unconditional with respect to (1.2) above?	VVM	46	The project is not developed under compliance markets. Therefore the checklist question is N/A.		OK	OK
1.4. Has(ve) the letter(s) of approval been issued by the respective Party's designated national authority (DNA)? If there is doubt with respect to (1.2) above, was verified with the DNA that the letter of approval is valid for the proposed CDM project activity under validation?	VVM	47	The project is not developed under compliance markets. Therefore the checklist question is N/A.		OK	OK
2. Participation						
2.1. Have all project participants been listed in a consistent manner in the project documentation?	VVM	51	The project is not developed under compliance markets. Therefore the checklist question is N/A.		OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
2.2. Is the information in tabular form of section A.3 consistent with the contact details provided in Annex 1 of the PDD?	VVM	52	The project is not developed under compliance markets. Therefore the checklist question is N/A.	OK	OK
2.3. Has the participation of each of the project participants been approved by at least one Party involved, either in a letter of approval or in a separate letter specifically to approve participation?	VVM	52	The project is not developed under compliance markets. Therefore the checklist question is N/A.	OK	OK
2.4. Are any entities other than those approved as project participants included in these sections of the PDD?	VVM	52	The project is not developed under compliance markets. Therefore the checklist question is N/A.	OK	OK
2.5. Has the approval of participation issued from the relevant DNA?	VVM	53	The project is not developed under compliance markets. Therefore the checklist question is N/A.	OK	OK
3. Project design document					
3.1. Is the PDD used as a basis for validation prepared in accordance with the latest template and guidance from the CDM Executive Board available on the UNFCCC CDM website?	VVM	55	The project is developed under the Voluntary Carbon Standard and uses the VCS PD template.	OK	OK
3.2. Is the PDD in accordance with the applicable CDM requirements for completing the PDD?	VVM	56	The checklist question is N/A as the VCS PD template is used. For compliance with the VCS PD template please see "Table 1. Validation requirements based on the Voluntary Carbon Standard 2007.1"	OK	OK
3.3. In CDM-PDD section A.1 -Title of project -Current version number and date of document	EB 41	Ann 12	The title and the current version and date of the document are given in section 1.1 of the VCS PD version 01.	OK	OK
3.4. In CDM-PDD section A.2, are following provided?	EB 41	Ann 12			
3.4.1. A brief description of the project activity covering purpose which includes the scenario existing prior to	EB 41	Ann 12	Under section 1.4 of the VCS PD version 01 a brief description of the project	CL12	OK

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the start of project, project scenario and baseline scenario	- VVM	- 58 59 60	activity is given covering the purpose of the project activity. Please also describe the scenario existing prior to the start of the project, project scenario and the baseline scenario under this section of the PD.		
3.4.2. Does the proposed CDM project activity involve the alteration of an existing installation or process?	VVM	63	The proposed project activity is a Greenfield investment.	OK	OK
3.4.3. Explanation on how the GHG emission reductions effected.	EB 41	Ann 12	Under section 1.4 of the VCS PD please give a brief description on how the proposed project activity reduces greenhouse gas emissions.	CL13	OK
3.4.4. The PP's views on the contribution of project activity to sustainable development	EB 41	Ann 12	PP's views on how the project activity contributes to sustainable development is given under section 1.16 of the VCS PD.	OK	OK
3.5. In CDM-PDD section A.3, are following provided in the tabular format? - List of project participants and parties - Identification of Host Party - Indication whether the Party wishes to be considered as project participant	EB 41 VVM	Ann 12 51,52	The related information on the project participants is given in a tabular form under section 1.15 of the VCS PD version 01.	OK	OK
3.6. In CDM-PDD section A.4.1, are following provided?	EB 41	Ann 12			
3.6.1. Physical description, location, host party(ies) and address as required	EB 41	Ann 12	The physical description of the project activity including the location is given under section 1.5 of the VCS PD version 01. The address of the PPs is given under section 1.15 of the VCS PD.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
3.6.2. Detailed physical location with unique identification of the project activity (e.g. Longitude/latitude)	EB 41	Ann 12	The detailed information on the project location, including geographic and physical information allowing the unique identification and delineation of the specific extent of the project are provided under section 1.5 of the VCS PD, however the following points needs further clarification: 1. Please provide the GPS coordinates separately for the power house and the reservoir. 2. Please also identify the grid connection point for the power plant. 3. Please clarify what the second arrow represents in Figure 2. "Project Location on Turkey Map"	CL01	OK
3.7. In CDM-PDD section A.4.2, is the list of categories of project activities provided?	EB 41	Ann 12	The category of the project activity is given under section 1.2 of the VCS PD version 01.	OK	OK
3.8. In CDM-PDD section A.4.3, are following provided?	EB 41	Ann 12			
3.8.1. A description of how environmentally safe and sound technology, and know-how, is transferred to the Host Party(ies)	EB 41	Ann 12	The checklist question is N/A as the VCS PD template is used. For compliance with the VCS PD template please see "Table 1. Validation requirements based on the Voluntary Carbon Standard 2007.1"	OK	OK
3.8.2. Further explanation of purpose of project activity with scenario existing prior to the start of project, scope or present activities and the baseline scenario	EB 41	Ann 12	The purpose of the project activity is given in section 1.4 of the VCS PD version 01.	OK	OK
3.8.3. List and arrangement of the main manufacturing/production technologies, systems and	EB 41	Ann 12	The equipment that will be used is given under section 1.9 of the PD version 01.	CL14	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
equipments involved			Please give a more detailed list and arrangement of the main manufacturing / production technologies, systems and equipments involved in the proposed project activity.		
3.8.4. The emissions sources and GHGs involved	EB 41	Ann 12	Under section 2.4 of the PD version 01, the GHG sources are identified for both the baseline scenario and the project activity. However in the table CH ₄ emissions from the reservoir has been considered twice. For the operation and construction of the HEPP CH ₄ emissions are considered to be negligible, but it is stated on the next row that CH ₄ emissions from reservoir is a main emission source. Also under this section the same information with the exact same sentences are given twice. Please clarify.	CL06	OK
3.9. In CDM-PDD section A.4.4, is the estimation of emission reductions provided as requested in a tabular format?	EB 41	Ann 12	The estimation of the emission reductions are given in a tabular format under section 1.3 of the VCS PD version 01.	OK	OK
3.10. In CDM-PDD section A.4.5, is information regarding public funding provided?	EB 41	Ann 12	Information about the public funding is given under section 1.4 of the VCS PD version 01.	OK	OK
3.11. In CDM-PDD section (Baseline identification)	EB 41	Ann 12			
3.11.1. The approved methodology and version number	EB 41	Ann 12	The project proponent has applied CDM approved Methodology ACM0002 version	CAR01	OK

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	VVM	69	10 and tools that are referred in ACM0002. However, ACM0002 version 10 is not the latest version of the methodology at the time of PD submission to the DOE. Please revise the PD using the latest version of the methodology.		
3.11.2. Are the following applicability conditions of the methodology ACM0002 v11 met?	VVM	70			
1. This methodology is applicable to grid-connected renewable power generation project activities that (a) install a new power plant at a site where no renewable power plant was operated prior to the implementation of the project activity (greenfield plants); (b) involve a capacity addition; (c) involve a retrofit of (an) existing plant(s); or (d) involve a replacement of (an) existing plant(s).	ACM	0002	The project is an installation a new hydropower plant where no renewable power plant was operated prior to the implementation of the project activity (greenfield plant). Hence, selected methodology and tools is convenient.	OK	OK
3.11.2.1. The project activity is the installation, capacity addition, retrofit or replacement of a power plant/unit of one of the following types: hydro power plant/unit (either with a run-of-river reservoir or an accumulation reservoir), wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit	ACM	0002	The project is an installation a new hydropower plant as validated through the site visit.	OK	OK
3.11.2.2. In case of hydro power plants, one of the following conditions must apply: - The project activity is implemented in an existing reservoir, with no change in the volume of reservoir; or	ACM	0002	The project activity results in construction of a new reservoir with a power density greater than 4 W/m ² .	OK	OK

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- The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the Project Emissions section, is greater than 4 W/m²; or - The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the Project Emissions section, is greater than 4 W/m².					
3.11.2.3. In the case of retrofits, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is "the continuation of the current situation, i.e. to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance".	ACM	0002	The project is an installation a new hydropower plant. Hence the question is N/A.	OK	OK
3.11.2.4. The methodology is not applicable to the following conditions. Please confirm • Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity • Biomass fired power plants; • Hydro power plants that result in new reservoirs or in the increase in existing reservoirs where the power density of the power plant is less than 4 W/m².	ACM	0002	It is confirmed that; • The project activity does not use fossil fuel. • Project does not consume biomass in any stage, as it is a hydropower plant. • The project activity results in a new reservoir.	OK	OK
3.12. Does the PDD correctly describe the project boundary, including the physical delineation of the proposed CDM project activity included within the project boundary for the purpose of calculating project and baseline	VVM	77 78	The PDD correctly describes the project boundary under section 2.4 of the PD version 01.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
emissions for the proposed CDM project activity?					
3.13. In CDM-PDD section B.3, are following provided? (a) Description of all sources and gases included in the project boundary in the table (b) A flow diagram of the project boundary physically delineating the project activity with all equipments, systems and flows of mass and energy etc	VVM EB 41	79 Ann 12	Under section 2.4 of the PD version 01, the GHG sources are identified for both the baseline scenario and the project activity. However in the table CH ₄ emissions from the reservoir has been considered twice. For the operation and construction of the HEPP CH ₄ emissions are considered to be negligible, but it is stated on the next row that CH ₄ emissions from reservoir is a main emission source. Also under this section the same information with the exact same sentences are given twice. Please clarify. The flow diagram of the project boundary is not a requirement of VCS.	CL06	OK
3.14. Is an explanation how the most plausible baseline scenario is identified in accordance with the selected baseline methodology is provided in CDM-PDD section B.4?	EB 41	Ann 12	The selected baseline scenario is the only option for the project activity. The baseline scenario is given under section 2.4 of the VCS PD version 01.	OK	OK
3.14.1. If the project activity is the install a new grid-connected renewable power plant/unit (greenfield plant), is the baseline scenario identified appropriately in accordance with the ACM0002 Ver.11?	ACM	0002	The baseline scenario is identified as follows: <i>"Electricity delivered to the grid by the project would have otherwise been generated by the operation of grid connected power plants and by the addition of new generation sources, as reflected in the Combined Margin calculations."</i>	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			The identified baseline scenario is in line with the methodology.		
3.14.2. If the project activity is a capacity addition to existing grid-connected renewable power plant/unit, is the baseline scenario identified appropriately in accordance with the ACM0002 Ver.10 and the point of time at which the generation facility would likely be replaced or retrofitted (DATE Baseline Retrofit) defined reasonably?	ACM	0002	The project activity is a Greenfield power plant, hence the checklist question is N/A.	OK	OK
3.14.3. If the project activity is the retrofit or replacement of existing grid-connected renewable power plant/unit, is the baseline scenario identified following step-wise procedure in accordance with the ACM0002 Ver.10?	ACM	0002	The project activity is a Greenfield power plant, hence the checklist question is N/A.	OK	OK
3.14.3.1. Are the realistic and credible alternative baseline scenarios for power generation appropriately identified following the Step 1 of the "Combined tool to identify the baseline scenario and demonstrate additionality"? (Step 1)	ACM	0002	The project activity is a Greenfield power plant, hence the checklist question is N/A.	OK	OK
3.14.3.2. Are the realistic and credible alternative baseline scenarios i.e. P1, P2 and P3 appropriately applied <i>Barrier analysis</i> following the Step 2 of the "Combined tool to identify the baseline scenario and demonstrate additionality"? (Step 2)	ACM	0002	The project activity is a Greenfield power plant, hence the checklist question is N/A.	OK	OK
3.14.3.3. If more than one alternative is remaining after Step 2, is <i>Investment analysis</i> appropriately applied (apply an Investment Comparison as per step 3 of the "Combined tool to identify the baseline scenario and demonstrate additionality" or a Benchmark Analysis as per step 2b of the "Tool for the demonstration and assessment of	ACM	0002	The project activity is a Greenfield power plant, hence the checklist question is N/A.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
additionality")? (Step 3)					
3.15. Does the PDD identify the baseline for the proposed CDM project activity, defined as the scenario that reasonably represents the anthropogenic emissions by sources of GHGs that would occur in the absence of the proposed CDM project activity?	VVM	80	The VCS PD identifies the baseline for the proposed CDM project activity, defined as the scenario that reasonably represents the anthropogenic emissions by sources of GHGs that would occur in the absence of the proposed CDM project activity, please also see checklist question 3.14.1 for the baseline scenario.	OK	OK
3.16. Has any procedure contained in the methodology to identify the most reasonable baseline scenario, been correctly applied?	VVM	81	The procedures have been applied correctly, as the project activity is a Greenfield Power Plant, there is only one applicable baseline scenario.	OK	OK
3.17. Does the selected methodology require use of tools (such as the "Tool for the demonstration and assessment of additionality" and the "Combined tool to identify the baseline scenario and demonstrate additionality") to establish the baseline scenario?	VVM	81	The selected methodology requires the use of tools "Tool for the demonstration and assessment of additionality" and "Tool to calculate the emission factor for an electricity system"	OK	OK
3.18. Does the methodology require several alternative scenarios to be considered in the identification of the most reasonable baseline scenario?	VVM	82	The project activity is a Greenfield Power Plant, there is only one applicable baseline scenario which is also identified by the project proponents under section 2.4 of the VCS PD version 01.	OK	OK
3.19. Are the documents and sources referred to in the PDD correctly quoted and interpreted And are they cross checked with other verifiable and credible sources, such as local expert opinion, if available? (identify the sources)	VVM	83	The documents and sources referred to in the VCS PD are correctly quoted and interpreted and they are cross checked with other verifiable and credible sources.	OK	OK
3.20. Have all applicable CDM requirements been taken into account in the identification of the baseline scenario	VVM	84	The project activity is developed under Voluntary Carbon Standard, therefore	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
for the proposed CDM project activity?			the checklist question is N/A.		
3.21. Have all relevant policies and circumstances been identified and correctly considered in the PDD, in accordance with the guidance by the CDM Executive Board?	VVM	84	The project activity is developed under Voluntary Carbon Standard, therefore the checklist question is N/A.	OK	OK
3.22. Does the PDD provide a verifiable description of the identified baseline scenario, including a description of the technology that would be employed and/or the activities that would take place in the absence of the proposed CDM project activity?	VVM	85	Under section 2.4 of the PD, after identifying the baseline scenario, please also include a description of the technology that would be employed and/or the activities that would take place in the absence of the proposed VCS project activity.	CAR04	OK
3.23. In CDM-PDD section B.5, are following provided?	EB 41	Ann 12			
3.23.1. Explanation and Justification of how and why this project activity is additional and therefore not the baseline scenario in accordance with the selected baseline methodology	EB 41	Ann 12	Explanation and justification of how and why this project activity is additional, therefore not the baseline scenario in accordance with the selected baseline methodology is given under section 2.5 of the VCS PD version 01.	OK	OK
3.23.2. Has the latest version of the "Tool for the demonstration and assessment of additionality" been used?	ACM	0002	The latest version (v05.2) of the "Tool for the demonstration and assessment of additionality" is used in the PD version 01.	OK	OK
3.23.3. Evidence that the incentive from the CDM was seriously considered in the decision to proceed with the project activity, if the starting date of the project activity is before the date of validation	EB 41	Ann 12	Under section 7 of the PD version 01 it is stated as follows: <i>"During the search of financing, the project owner was aware of that GHG emission trade could be an option to straighten the feasibility of the project"</i> The prior consideration of the VER	OK	OK

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			benefits was proven with a board decision No: 2007/8 dated 14.09.2007.		
3.24. In CDM-PDD section B.6.1, are following provided? (Algorithms and/or formulae used to determine emission reductions)	EB 41	Ann 12			
3.24.1. Explanation how the procedures, in the approved methodology to calculate project emissions, baseline emissions, leakage emissions and emission reductions are applied to the proposed project activity	EB 41	Ann 12	The explanations on how the procedures in the approved methodology are applied to the project activity are given under section 2.4 of the VCS PD version 01.	OK	OK
3.24.2. Do the steps taken and equations applied to calculate project emissions, baseline emissions, leakage and emission reductions comply with the requirements of the selected baseline and monitoring methodology?	VVM	88	1. Under section 2.4 of the PD version 01, Baseline emissions formula is given as: $BE_y = (EG_y - EG_{baseline}) \times EF_{Grid, CM, y}$ However this is not the formula given in the methodology for Greenfield power plants. Although the results will not change, please revise the PD so that the correct formulas are included in the PD. Also under the same section, the value of EG_y is given as 100,646 MWh/year however in the baseline calculations, the grid emission factor is multiplied by 156,205 MWh/year. Please clarify. 2. Under section 1.3 of the PD version 01, although the power density is calculated to be greater than 10 W/m² the Project Emission formula is given after this calculation (the formula is given twice under this section) Also it is stated that the	CL15	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			annual $PE_{HP,y}$ is given in Table 1 which does not exist in this section. Please clarify.		
3.24.2.1. Are the Project emissions appropriately calculated?	ACM	0002	The project emissions are calculated under sections 1.3 and 2.4 of the PD version 01. Please clarify and correct the following points regarding the power density calculations: 1. Under section 1.3, the value of Cap_{PJ} is given as 84,000,000 W. Under section 2.4, the value of Cap_{PJ} is given as 20,500,000 W, However the installed capacity of the project was defined as 63 MW in other sections of the PD version 01. 2. Under section 1.3, the value of A_{PJ} is given as 13.4 km², under section 2.4 of the PD the value of A_{PJ} is given as 13,837 m². Please clarify where this value is taken from and which value is the correct one by giving references. 3. Please revise the Power Density calculations using the correct values, making sure that the calculations and the data used in the calculations are the same throughout the whole PD.	CAR05	OK
3.24.2.2. Are the Baseline emissions appropriately calculated specifically for (a)greenfield plants or (b) retrofit and replacements or (c) capacity additions?	ACM	0002	The baseline emissions are appropriately calculated for the project activity taking into consideration that it is a Greenfield investment.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
3.24.2.3. Are the Leakage appropriately calculated?	ACM	0002	Leakage is considered to be negligible by the methodology.	OK	OK
3.24.2.4. Are the Emission reductions appropriately calculated?	ACM	0002	The emission reductions are appropriately calculated.	OK	OK
3.24.3. Have the equations and parameters in the PDD been correctly applied with respect those in the select approved methodology?	VVM	89	The correctness of the application of the equations and parameters are assessed in detail in questions 3.24.5-6-7-8 of this checklist.	OK	OK
3.24.4. Does the methodology provide for selection between different options for equations or parameters?	VVM	89	Yes, the methodology and the referred tool "Tool to calculate the emission factor for an electricity system" provides a selection between different options for equations or parameters.	OK	OK
3.24.5. If yes, has adequate justification been provided and correct equations and parameters been used in accordance with the methodology selected?	VVM	89	The justifications about the selections have been given in the PD version 01. However the following issues need correction: 1. Under section 2.4, Step 2 of the PD version 01, it is stated that "option C of Simple OM method was adopted. However "tool to calculate emission factor for an electricity system" does not have any option C for the Simple OM method. Please clearly identify the method chosen under this step. 2. Under section 2.4 step 2, it is further stated that "<i>Simple OM Option C Method is eligible when low cost and/or must run resources constitute as an average of the five</i>	CAR06	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			<i>most recent years, less than 50% of the total generation of the grid"</i> However in the tool this prerequisite is given for all of the options of Simple OM and it is stated as follows: <i>"The simple OM method (option a) can only be used if low-cost/must-run resources constitute less than 50% of total grid generation in: 1) average of the five most recent years, or 2) based on long-term averages for hydroelectricity production."</i> Please justify why the average of five most recent years is calculated. The tool suggests calculating the share of LCMR based on long term averages. 3. In the PD version 01 under the section 2.4 Step 4, Simple OM Option B is used however there is no justification regarding this choice. According to the Tool there are some applicability conditions regarding the Simple OM option B, please justify your choice taking into consideration the conditions set out in the tool. 4. In the baseline calculation excel sheet, it is seen that the value of NCV of fuels is calculated using data from TEIAS, however under section 2.4 step 4, it is not		

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			explained and justified how the NCV is calculated. 5. Under section 2.4 Step 5, there are two options namely Option 1 and Option 2, and according to the tool the option chosen shall be documented in the PD. Please document and justify your choice under Step 5. 6. Under section 2.4 Step 6 the formula $EF_{grid,BM,y}$ is given. According to the tool <i>"The CO2 emission factor of each power unit m ($EF_{EL,m,y}$) should be determined as per the guidance in Step 4 (a) for the simple OM, using options A1, A2 or A3, using for y the most recent historical year for which power generation data is available, and using for m the power units included in the build margin."</i> However it is not clear in the PD which equation is chosen for the calculation of the $EF_{EL,m,y}$. Please clearly identify which equation is used and justify this selection.		
3.24.6. If yes, have correct equations and parameters been used, in accordance with the methodology selected?	VVM	89	The equations and parameters used in the PD are correct in general however under the section 2.4 Step 4, the formula of the $EF_{grid,OMsimple,y}$ is given however the explanation of this formula does not match the explanation given in the tool. Please revise the PD to include	CL16	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			the correct explanations.		
3.24.7. Appropriate and correct?	VVM	90	Please see checklist questions 3.24.2, 3.24.5 and 3.24.6	OK	OK
3.24.8. Applicable to the proposed CDM project activity?	VVM	90	The equations and parameters used in the PD version 01 are all applicable to the proposed project activity.	OK	OK
3.24.9. Resulting in a conservative estimate of the emission reductions?	VVM	90	The equations and parameters used in the PD version 01 results in a conservative estimate of emission reductions.	OK	OK
3.24.10. A compilation of information on the data and parameters that are not monitored throughout the crediting period but that are determined only once and thus remains fixed throughout the crediting period and that are available when validation is undertaken	EB 41	Ann 12	In the PD version 01, a compilation of information on the data and parameters that are not monitored throughout the crediting period but that are determined only once and thus remains fixed throughout the crediting period and that are available when validation is undertaken is given. Some of these parameters (namely $FC_{i,y}$ and $NCV_{i,y}$) are also included in the monitoring plan, and the monitoring period of these parameters are given as "Once for each crediting period" as suggested by the tool.	OK	OK
3.24.11. Explanation and justification for the choice of the source of data	EB 41	Ann 12	The choice of the source of data is explained and justified in the PD version 01.	OK	OK
3.24.12. Clear and transparent references or additional documentation in Annex 3	EB 41	Ann 12	The references of the data are given clearly in the PD version 01. Additional information about the calculations are given in detail in	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			Annexes 3,4 and 6.		
3.24.13. Where values have been measured, a description of the measurement methods and procedures (e.g. which standards have been used), indicated the responsible person/entity having undertaken the measurement, the date of measurement(s) and the measurement results	EB 41	Ann 12	The values used in the calculations are not measured.	OK	OK
3.25. In CDM-PDD section B.6.3, are following provided?	EB 41	Ann 12			
3.25.1. A transparent ex ante calculation of project emissions, baseline emissions (or, where applicable, direct calculation of emission reductions) and leakage emissions expected during the crediting period, applying all relevant equations provided in the approved methodology	EB 41	Ann 12	The ex-ante calculation of emission reductions are given under section 2.4 of the VCS PD version 01.	OK	OK
3.25.2. Documentation how each equation is applied, in a manner that enables the reader to reproduce the calculation	EB 41	Ann 12	Please give more detailed information on how each equation is applied to enable the reader to reproduce the calculations. For example, under section 2.4, step 7: 1. For the calculation of the $EF_{grid,CM,y}$ when the values given in the PD are used in the formula, the result is 0,6135 however this value is taken as 0,614 in the calculations. 2. Under the same section for the calculation of BE_y when the values given are used in the formula the result is 95,909.87 tCO_{2e}, however the result is given as 95,927 tCO_{2e} in the PD version 01. Please clarify.	CL17	OK
3.25.3. Additional background information and or data in	EB 41	Ann	The data given on the spreadsheets are	OK	OK

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Annex 3, including relevant electronic files (i.e. spreadsheets)		12	also given in Annexes 3, 4 and 6 of the PD version 01.		
3.26. In CDM-PDD section B.6.4 are, the results of the ex ante estimation of emission reductions for all years of the crediting period, provided in a tabular format?	EB 41	Ann 12	A table showing the results of the ex ante estimation of emission reductions for all years of the crediting period is given in section 1.3 of the PD version 01.	OK	OK
3.27. In CDM-PDD section B.7.1, are following provided?	EB 41	Ann 12			
3.27.1. Specific information on how the data and parameters that need to be monitored would actually be collected during monitoring for the project activity	EB 41	Ann 12	Under section 3.2 of the VCS PD version 01, specific information on how the data and parameters that need to be monitored would actually be collected during the monitoring period is given.	OK	OK
3.27.2. For each parameter the following below information, using the table provided:	EB 41	Ann 12			
3.27.2.1. The source(s) of data that will be actually used for the proposed project activity (e.g. which exact national statistics). Where several sources may be used, explain and justify which data sources should be preferred.	EB 41	Ann 12	The origin of the data is given under section 3.2 of the VCS PD version 01 in a tabular format. The source of data is also given under section 3.3 of the VCS PD version 01 within the data/parameter tables. For the parameter $EG_{\text{facility},y}$ the source of data is given as "metering device from the grid operator" under section 3.2 and as "monthly invoice" under section 3.3. In the methodology the source of data for this parameter is defined as "Project activity site" Please define the same source for the same parameter and make sure the source is in line with the methodology.	CAR03	OK

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3.27.2.2. Where data or parameters are supposed to be measured, specify the measurement methods and procedures, how the measurement is undertaken: (i) A description of the QA/QC procedures (if any) that should be applied; (ii) Where relevant: any further comment.	EB 41	Ann 12	Measurement procedures for each data/parameter is given in the tabular format, however for the parameter EG _{facility,y} the measurement procedure is given as MWh. Please define the measurement procedures as per the methodology.	CL09	OK
3.28. In CDM-PDD section B.7.2, is a detailed description of the monitoring plan provided?	EB 41	Ann 12	Detailed description of the monitoring plan is given under sections 3.2 and 3.4 of the VCS PD version 01.	OK	OK
3.29. Are all data monitored as per monitoring methodology?	ACM	0002	All the data and parameters are monitored as per the monitoring methodology.	OK	OK
3.30. Are all data collected as part of monitoring archived electronically and kept at least for 2 years after the end of the last crediting period?	ACM	0002	Under section 3.1 of the VCS PD version 01 it is stated as follows: <i>"The monitoring methodology is applicable as the data will be archived electronically and be kept at least for 2 years after the end of the last crediting period."</i>	OK	OK
3.31. In CDM-PDD section B.8, are following provided?	EB 41	Ann 12			
3.31.1. Date of completion of the application of the methodology to the project activity study in DD/MM/YYYY	EB 41	Ann 12	Under section 2.1 of the VCS PD version 01, it is stated that the completion date of the baseline study is 06/10/2009, however the date of the first version of the PD is March 16 2010. Please clarify.	CL18	OK
3.31.2. Contact information of the person(s)/entity(ies) responsible for the application of the baseline and	EB 41	Ann 12	The contact information of the person(s)/entity(ies) responsible for the application of the baseline and	OK	OK

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monitoring methodology to the project activity			monitoring methodology to the project activity is given under section 3.4 of the VCS PD version 01.		
3.31.3. Indication if the person/entity is also a project participant listed in Annex 1	EB 41	Ann 12	The project is developed under VCS hence the checklist question is N/A.	OK	OK
3.32. In CDM-PDD section C.1.1, are following provided?	EB 41	Ann 12			
3.32.1. Is the project's starting date clearly defined and evidenced?	EB 41	Ann 12	The project starting date is defined as 1 st of July 2010 under section 1.6 of the VCS PD version 01.	OK	OK
3.33. In CDM-PDD section D., are the conclusions and all references to support documentation of an environmental impact assessment undertaken in accordance with the procedures as required by the Host Party, if environmental impacts are considered significant by the project participants or the Host, provided?	EB 41	Ann 12	A summary of the environmental impacts is given under section 5 of the VCS PD version 01. However in this section it is stated that the domestic law and regulation does not require any environmental impact assessment. Please clarify as this is not the case at the moment. Please also include a summary of the EIA prepared by the project owner which was also sent to the validation team.	CL11	OK
3.34. In CDM-PDD section E.1, are the following provided?	EB 41	Ann 12			
3.34.1. The process by which comments by local stakeholders have been invited and compiled. An invitation for comments by local stakeholders shall be made in an open and transparent manner, in a way that facilitates comments to be received from local stakeholders and allows for a reasonable time for comments to be submitted.	EB 41	Ann 12	Under section 6 of the VCS PD it is stated as follows: <i>"The project owner used different channels to invite comments by stakeholders. Most of the meetings were held through the Mukhtar (village governors) and Council of Elderly (Ihtiyar Heyeti; the village council) of</i>	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			<i>Cataldere village, and the officials of Municipality of Kaptanpasa. The project owner visited various households in the project area to comprehend their needs and to identify how the project could assist in meeting the social and economic needs of the community."</i>		
3.34.2. The project activity is described in a manner, which allows the local stakeholders to understand the project activity, taking into account confidentiality provisions of the CDM modalities and procedures.	EB 41	Ann 12	Under section 6 of the VCS PD version 01, it is stated as follows: "Although the domestic laws and regulations do not require any stakeholders' participation, both in the project planning and construction phase, the project owner managed to conduct various unofficial and unstructured stakeholder participation meetings." Please clarify which domestic law is referred here and how this law applies to the project activity. Please also describe in detail how the project activity is described to the stakeholders in these meetings.	CL19	OK
3.34.3. The local stakeholder process has been, completed before submitting the proposed project activity to the DOE for validation.	EB 41	Ann 12	Please clarify when the stakeholder consultation process is finalized.	CL20	OK
3.35. In CDM-PDD section E.2, are following provided?	EB 41	Ann 12			
3.35.1. Identification of local stakeholders that have made comments	EB 41	Ann 12	Please identify the names of the stakeholders that have made comments.	CL21	OK
3.35.2. A summary of these comments.	EB 41	Ann 12	A summary of the comments are given under section 6 of the VCS PD version	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			01.		
3.36. In CDM-PDD section E.3 is the explanation of how due account have been taken of comments received from local stakeholders provided?	EB 41	Ann 12	How due account has been taken of comments received from the stakeholders is given under section 6 of the VCS PD version 01.	OK	OK
3.37. In CDM-PDD Annex 1, are the following provided?	EB 41	Ann 12			
3.37.1. Contact information of project participants	EB 41	Ann 12	The contact information of the project participants is given under section 1.15 of the PD version 01.	OK	OK
3.37.2. For each organization listed in section A.3 the following mandatory fields: Organization, Name of contact person, Street, City, Postfix/ZIP, Country, Telephone and Fax or e-mail	EB 41	Ann 12	For each organization listed, the name of contact person, street, city, zipcode, country, telephone, fax and e-mail is given under section 1.15 of the PD version 01.	OK	OK
3.38. In CDM-PDD Annex 2, is information from Parties included in Annex I on sources of public funding for the project activity which shall provide an affirmation that such funding does not result in a diversion of official development assistance and is separate from and is not counted towards the financial obligations of those Parties provided?	EB 41	Ann 12	The project activity is developed under Voluntary Carbon Standard, therefore the checklist question is N/A.	OK	OK
3.39. In CDM-PDD Annex 3, is the background information used in the application of the baseline methodology provided?	EB 41	Ann 12	The background information about the baseline study is given in Annexes 3, 4 and 6 of the VCS PD version 01.	OK	OK
3.40. In CDM-PDD Annex 4, is the background information used in the application of the monitoring methodology provided?	EB 41	Ann 12	The project activity is developed under Voluntary Carbon Standard, therefore the checklist question is N/A.	OK	OK
4. Additionality of a project activity					
4.1. General checklist for additionality					

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
4.1.1. Does the CDM-PDD state the latest version of the additionality tool being used?	VVM	94	Under section 2.5 of the VCS PD version 01 it is stated that version 05.2 of the "Tool for the demonstration and assessment of additionality" which is the latest version.	OK	OK
4.1.2. Were the steps taken of the "Tool for the Demonstration and Assessment of Additionality" to assess additionality used:	EB 39	Ann 10	All of the steps defined in the additionality tool are used in PD version 01.	OK	OK
4.1.3. Have the following alternatives been included while defining alternatives as per sub-step 1a?	EB 39	Ann 10			
4.1.3.1. The proposed project activity undertaken without being registered as a CDM project activity;	EB 39	Ann 10	"The proposed project activity not undertaken as a VER project activity" is defined as Alternative 2, under section 2.5 of the VCS PD version 01.	OK	OK
4.1.3.2. Other realistic and credible alternative scenario(s) to the proposed CDM project activity scenario that deliver outputs services or services with comparable quality, properties and application areas, taking into account, where relevant, examples of scenarios identified in the underlying methodology;	EB 39	Ann 10	"Construction of a thermal (natural gas) power plant with the same annual power output" is defined as Alternative 1. "Construction of some other renewable energy plant with the same annual power output" is defined as Alternative 3, under section 2.5 of the VCS PD version 01.	OK	OK
4.1.3.3. If applicable, continuation of the current situation (no project activity or other alternatives undertaken).	EB 39	Ann 10	"The same service of power supply is provided from the grid (Uzunder I HEPP is not built)" is defined as Alternative 4, under section 2.5 of the VCS PD version 01.	OK	OK
4.1.4. Has the outcome of Step 1a: Identified realistic and credible alternative scenario(s) to the project activity done correctly?	EB 39	Ann 10	The outcome of Sub-step 1a is identified realistic and credible alternatives which are alternatives 1, 2, and 4.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			Alternative 3 is eliminated because of the lack of resources in the region.		
4.1.5. Is the alternative(s) in compliance with all mandatory applicable legal and regulatory requirements, even if these laws and regulations have objectives other than GHG reductions, e.g. to mitigate local air pollution, and outcome of Step 1.b is thus concluded?	EB 39	Ann 10	Under sub-step 1b the relevant laws are identified. All of the alternatives are in compliance of the identified laws as these alternatives are the widely operational in the country.	OK	OK
4.1.6. If an alternative does not comply with all mandatory applicable legislation and regulations, has it been shown that, based on an examination of current practice in the country or region in which the law or regulation applies, those applicable legal or regulatory requirements are systematically not enforced and that noncompliance with those requirements is widespread in the country?	EB 39	Ann 10	N/A	OK	OK
4.1.7. Has PP selected Step 2 (Investment analysis) or Step 3 (Barrier analysis) or both Steps 2 and 3?	EB 39	Ann 10	The PP has selected Step 2 Investment Analysis and Step 3 Barrier Analysis.	OK	OK
4.1.8. In step 2, have all the sub-steps as below been followed?	EB 39	Ann 10	All the sub-steps of Step 2 has been followed.	OK	OK
4.1.9. In sub-step 2a has the determination of appropriate method of analysis done as per the guidance as below?	EB 39	Ann 10	The Benchmark Analysis is assessed to be appropriate under Sub-Step 2a.	OK	OK
4.1.9.1. Simple cost analysis if the CDM project activity and the alternatives identified in Step 1 generate no financial or economic benefits other than CDM related income (Option I).	EB 39	Ann 10	N/A	OK	OK
4.1.9.2. Otherwise, use the investment comparison analysis (Option II) or the benchmark analysis (Option III). Specify option used with justification.	EB 39	Ann 10	The Benchmark Analysis (Option III) is selected however it is not justified why benchmark analysis is selected.	CL22	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			Please clarify.		
4.1.10. Has the below guideline followed for sub-step 2b Option I. Apply simple cost analysis? Document the costs associated with the CDM project activity and the alternatives identified in Step1 and demonstrate that there is at least one alternative which is less costly than the project activity.	EB 39	Ann 10	Simple cost analysis is not used, hence the checklist question is N/A.	OK	OK
4.1.11. Has the below guideline followed for sub-step 2b Option II. Apply investment comparison analysis? Identify the financial indicator, such as IRR, NPV, cost benefit ratio, or unit cost of service most suitable for the project type and decision-making context. Please specify	EB 39	Ann 10	Sub-Step 2b, Option II is not used, instead Option III is selected. The financial indicator is identified as Equity IRR under Sub-Step 2b.	OK	OK
4.1.12. Has the most suitable benchmark for the project been determined in Sub-step 2b?	EB 39	Ann 10	The selected benchmark is suitable for the type of IRR calculated.	OK	OK
4.1.12.1. Which source shall the discount rates and benchmarks derived from? Please specify benchmark and justify.	EB 39	Ann 10	The benchmark selected is given as follows under Sub-Step 2b: <i>"Almost all bankers who have been asked for an opinion for the minimum IRR that is required for the approval of a similar project expressed some after tax IRR that is in the bandwidth of 16 to 18 percent."</i> This type of benchmark is suitable for the type of IRR calculated, however please send objective evidence to prove this statement. Please be informed that at any stage of this validation, the proof of this benchmark can be requested to be made public from the PP.	CL23	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
4.1.13. Has the below guideline followed for Sub-step 2c: Calculation and comparison of financial indicators (only applicable to Options II and III)?	EB 39	Ann 10			
4.1.13.1. Calculate the suitable financial indicator for the proposed CDM project activity and, in the case of Option II above, for the other alternatives. Include all relevant costs (including, for example, the investment cost, the operations and maintenance costs), and revenues (excluding CER revenues, but possibly including inter alia subsidies/fiscal incentives, ODA, etc, where applicable), and, as appropriate, non-market cost and benefits in the case of public investors if this is standard practice for the selection of public investments in the host country.	EB 39	Ann 10	The financial indicator is selected as equity IRR and Option III Benchmark analysis is selected. The equity IRR is a suitable financial indicator for this type of project activities, and IRR is calculated and details are shown both in the PD version 01 and the excel calculation sheet.	OK	OK
4.1.13.2. Present the investment analysis in a transparent manner and provide all the relevant assumptions, preferably in the CDM-PDD, or in separate annexes to the CDM-PDD.	EB 39	Ann 10	The investment analysis is presented clearly in the PD version 01. The details of the investment analysis is given in a separate excel calculation sheet which was reviewed by the financial expert.	OK	OK
4.1.13.3. Justify and/or cite assumptions.	EB 39	Ann 10	The assumptions have been cited in the VCS PD version 01.	OK	OK
4.1.13.4. In calculating the financial/economic indicator, the project's risks can be included through the cash flow pattern, subject to project-specific expectations and assumptions.	EB 39	Ann 10	The calculation of the financial indicator is done using the data from the FSR and all the values used in the calculation including the risks are taken from this report.	OK	OK
4.1.13.5. Assumptions and input data for the investment analysis shall not differ across the project activity and its alternatives, unless differences can be well substantiated.	EB 39	Ann 10	Benchmark analysis is applied hence the checklist question is N/A.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
4.1.13.6. Present in the CDM-PDD a clear comparison of the financial indicator for the proposed CDM activity. Please specify details for above.	EB 39	Ann 10	The financial indicator is compared to a benchmark. The comparison is made in a clear and transparent manner. However benchmark shall be clarified. Please see CL23.	OK	OK
4.1.13.7. Is the period of assessment limited to the proposed crediting period of the CDM project activity?	EB 41	Ann 45	The period of assessment is limited to the license duration which is 49 years which is more than the crediting period.	OK	OK
4.1.13.8. Does the project IRR and equity IRR calculations reflect the period of expected operation of the underlying project activity (technical lifetime), or - if a shorter period is chosen - include the fair value of the project activity assets at the end of the assessment period?	EB 41	Ann 45	The equity IRR calculations reflect the license duration as an assessment period. However the period of assessment shall be limited to the lifetime of the turbines (operational lifetime). Please revise the assessment period taking into consideration that the turbines are manufactured by Alstom.	CAR07	OK
4.1.13.9. Does the IRR calculation include the cost of major maintenance and/or rehabilitation if these are expected to be incurred during the period of assessment?	EB 41	Ann 45	The IRR calculations includes the major maintenance costs.	OK	OK
4.1.13.10. Do the project participants justify the appropriateness of the period of assessment in the context of the underlying project activity, without reference to the proposed CDM crediting period?	EB 41	Ann 45	The financial analysis is performed over a 49 year period which is the license duration of the project activity. This is a conservative assumption as it covers not only the costs along the crediting period but also the costs along the estimated lifetime of the proposed project activity. However to be more realistic the assessment shall be limited to the lifetime of the turbines. Please see CAR 07.	CAR07	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
4.1.13.11. Does the cash flow in the final year include a fair value of the project activity assets at the end of the assessment period?	EB 41	Ann 45	The cash flow in the final year does not include a fair value of the project assets because the assessment period covers the project lifetime.	OK	OK
4.1.14. Has the below guideline followed for Sub-step 2d: Sensitivity analysis (only applicable to Options II and III)? Include a sensitivity analysis that shows whether the conclusion regarding the financial/economic attractiveness is robust to reasonable variations in the critical assumptions.	EB 39	Ann 10	A sensitivity analysis is included to show whether the conclusions regarding the financial/economic attractiveness is robust to reasonable variations in the critical assumptions.	OK	OK
4.1.15. Has the outcome of Step 2 clearly mentioned with justification?	EB 39	Ann 10	The outcome of Step 2 is clearly mentioned as follows in the PD version 01: <i>"The above sensitivity analysis reveals the fact that no alternative scenario without VER revenues can make the project pass the benchmark IRR expectation which is in 16% to 18% range. Therefore, the project is not financially attractive."</i>	OK	OK
4.1.16. Have the barrier analysis been conducted?	EB 39	Ann 10	The barrier analysis is included in the PD version 01. The barriers were identified under two main categories, the names of the categories and the barriers that are identified under these categories is given below: 1. Investment Barriers - Increases in the exchange rate - Price cap for renewable electricity - Market liberalization and privatization of electricity	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			distribution • Availability of debt financing and global recession 2. Technological Barriers • Technical Innovation • Planning problems with low scale maps • Harsh geological conditions • Lack of adequate land registry in the region		
4.1.17. In step 4: Common practice analysis has all the sub-steps as below followed?	EB 39	Ann 10			
4.1.17.1. Has the below guideline followed for Sub-step 4a: Analyze other activities similar to the proposed project activity? Provide an analysis of any other activities that are operational and that are similar to the proposed project activity. Other CDM project activities are not to be included in this analysis. Provide documented evidence and, where relevant, quantitative information. On the basis of that analysis, describe whether and to which extent similar activities have already diffused in the relevant region.	EB 39	Ann 10	Under sub-step 4a of the common practice analysis the activities similar to the proposed project activities are analyzed. First this analysis is made for the Eastern Black Sea Region and on the basis of the installed capacity. It is seen with this analysis that the project activities that are present in the region are not yet operational and have very low completion rates (below 10%). When this analysis was extended to the host country, it was seen that the participation of private sector in electricity production from hydropower is a very new concept and only 57 projects out of 477 were completed more than 50% on the date of preparation of the PDD. In addition, the new capacity from	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			privately owned hydro facilities represents a friction of new facilities added to Turkey's capacity every year. On average, the new privately owned hydro facilities represents only 2.3% of new additions. Similarly, new additions from privately held hydro projects represents only 0,1% of Turkey's installed capacity in the last 6 years.		
4.1.17.2. Has the below guideline followed for Sub-step 4b: Discuss any similar Options that are occurring?	EB 39	Ann 10	Under sub-step 4b it is stated as follows: <i>"There is no accessible information on the projects benefit from carbon revenues. Therefore, it is not possible to compare the projects with the subject project in respect to carbon revenues. However, the low rate of completion of the projects and the low contribution privately held hydro projects confirm the barriers elaborated above and also proves that the electricity generation from HEPP business is not a common practice."</i> The explanation is found satisfactory by the validation team.	OK	OK
4.1.18. Has the outcome from Step 4 clearly mentioned in PDD?	EB 39	Ann 10	Outcome of step 4 is given as <i>"There is no accessible information on the projects benefit from carbon revenues. Therefore, it is not possible to compare the projects with the subject project in respect to carbon revenues. However, the low rate of completion of the projects and the low contribution</i>	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			<i>privately held hydro projects confirm the barriers elaborated above and also proves that the electricity generation from HEPP business is not a common practice. Therefore Step 4 is satisfied".</i>		
4.2. Prior consideration of the clean development mechanism					
4.2.1. Is the project activity start date prior to the date of publication of the PDD for stakeholder comments?	VVM	96	The project activity is developed under VCS, hence the PD is not published for stakeholder comments. The checklist question is N/A.	OK	OK
4.2.2. If yes, were the CDM benefits considered necessary in the decision to undertake the project as a proposed CDM project activity?	VVM	96	N/A	OK	OK
4.2.3. Is the start date of the project activity, reported in the PDD, in accordance with the "Glossary of CDM terms", which states that "The starting date of a CDM project activity is the earliest date at which either the implementation or construction or real action of a project activity begins"?	VVM	97	The project activity is developed under VCS and the definition of "Project Start Date" is different in VCS 2007.1. The checklist question is N/A.	OK	OK
4.2.4. Does the project activity require construction, retrofit or other modifications?	VVM	97	The project activity is developed under VCS and the definition of "Project Start Date" is different in VCS 2007.1. The checklist question is N/A.	OK	OK
4.2.5. Is it ensured that the date of commissioning cannot be considered as the project activity start date?	VVM	97	The project activity is developed under VCS and the definition of "Project Start Date" is different in VCS 2007.1. The checklist question is N/A.	OK	OK
4.2.6. Is it a new project activity (project activities with starting date on or after 02 August 2008) or an	VVM	98	The project activity is developed under VCS and the definition of "Project Start	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
existing project activity (project activities with a start date before 02 August 2008)?			Date" is different in VCS 2007.1. The checklist question is N/A.		
4.2.7. For a new project, for which PDD has not been published for global stakeholder consultation or a new methodology proposed to the Executive Board before the project activity start date, had the PP informed the Host Party DNA and/or the UNFCCC secretariat in writing of the commencement of the project activity and of their intention to seek CDM status?	VVM	99	The project activity is developed under VCS, hence the checklist question is N/A.	OK	OK
4.2.8. For an existing project activity, for which the start date is prior to the date of publication of the PDD for global stakeholder consultation, are the following evidences provided:	VVM	100	The project activity is developed under VCS, hence the checklist question is N/A.	OK	OK
4.2.8.1. Evidence that must indicate that awareness of the CDM prior to the project activity start date, and that the benefits of the CDM were a decisive factor in the decision to proceed with the project,	VVM	100	Under section 7 of the PD version 01 it is stated as follows: <i>"During the search of financing, the project owner was aware of that GHG emission trade could be an option to straighten the feasibility of the project"</i> The prior consideration of the VER benefits was proven with a board decision No: 2007/8 dated 14.09.2007.	OK	OK
4.2.8.2. Reliable evidence from project participants that must indicate that continuing and real actions were taken to secure CDM status for the project in parallel with its implementation	VVM	100	The prior consideration of the VER benefits was proven with a board decision No: 2007/8 dated 14.09.2007.	OK	OK
4.3. Identification of alternatives					
4.3.1. Does the approved methodology that is selected by the proposed CDM project activity prescribe the baseline scenario and hence no further analysis is	VVM	103	The applied methodology prescribes the baseline scenario; hence no further	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
required?			analysis is required.		
4.3.2. If no, does the PDD identify credible alternatives to the project activity in order to determine the most realistic baseline scenario?	VVM	103	N/A	OK	OK
4.3.3. Does the list of alternatives given in the PDD ensure that: - One of the options that the project activity is undertaken without being registered as a proposed CDM project activity - The list contains all plausible alternatives - The alternatives comply with all applicable and enforced legislation	VVM	104	N/A	OK	OK
4.4. Investment analysis					
4.4.1. If investment analysis has been used to demonstrate the additionality of the proposed CDM project activity , does the PDD provide evidence that the proposed CDM project activity would not be:	VVM	106			
4.4.1.1. The most economically or financially attractive alternative?	VVM	106	The project uses benchmark analysis, therefore the checklist question is N/A.	OK	OK
4.4.1.2. Economically or financially feasible, without the revenue from the sale of certified emission reductions (CERs)?	VVM	106	The project uses benchmark analysis and the PPs are trying to prove that the project is not economically or financially feasible without the VER revenues.	OK	OK
4.4.2. Was this shown by one of the following approaches?	VVM	107			
4.4.2.1. Demonstrate that the proposed CDM project activity would produce no financial or economic benefits other than CDM-related income.	VVM	107	The project uses benchmark analysis, therefore the checklist question is N/A.	OK	OK
4.4.2.2. The proposed CDM project activity is less economically or financially attractive than at least	VVM	107	The project uses benchmark analysis, therefore the checklist question is N/A.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
one other credible and realistic alternative.					
4.4.2.3. The financial returns of the proposed CDM project activity would be insufficient to justify the required investment.	VVM	107	It is shown in the PD version 01 that the financial returns of the proposed VCS project activity would be insufficient to justify the required investment.	OK	OK
4.4.3. Was a thorough assessment of all parameters and assumptions used in calculating the relevant financial indicator, and determine the accuracy and suitability of these parameters using the available evidence and expertise in relevant accounting practices conducted?	VVM	109	The following issues needs to be clarified regarding the investment analysis section of the PD version 01 and the excel calculation sheet named "Karadeniz Uzundere I_IRR.xls": 1. Please be in line with the statutory accounting principles regarding the tax income. For example, please start the income tax application after the cumulative income becomes positive. 2. In the cash flow scenarios in which VER Revenue is included, there must be income tax charged for the additional income of VER. In the analysis, VER income is added to the original cash flow directly. Please, apply income tax for the VER incomes. 3. Please provide the finance expense figures in the analysis supported by the formulas. It seems very high; 98.8 k \$. That's why the IRR figure is very low. Please note that, the total amount received is 73.8m \$. 4. In relation with this comment, it is seen that there is an interest expense in 2009 and creates a negative cash	CAR08	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			flow at the end of year. Please clarify how this negative balance will be compensated. If it is thought that, this negative balance will be compensated by the shareholders' equity, then change the loan/equity ratio accordingly. 5. Please present whether VAT (Value Addet Tax – KDV) is included in the investment amount of 92.7m \$ or not? If it is included, it should be deducted in the following years. 6. In the analysis, it is seen that asset amortization starts before the construction period is finished. It would be beter to start the amortization after the installation is fully completed based on the statutory accounting principles. 7. Please correct the depreciation periods. Construction must be 40 years and electromechanical equipment must be 30 years. 8. Please verify that there is no license fee in the investment. 9. Please verify if the Debt Repayments represented in the IRR analysis are the Capital Repayments. If it is so, the total debt repayment amount (50.1m \$) is not equal to the Debt figure (73.7m \$) in row 124 in the analysis. Please correct it.		

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			10. Please correct the depreciation periods at first, and then include the fair value of the project as a cash inflow at the end of the assessment period. 11. In the PDD 10% of contingency is added to the Project investment but in the IRR analysis it is not taken into consideration. Please be in line with the PDD.		
4.4.4. Was the sensitivity analysis by the project participants to determine under what conditions variations in the result would occur and the likelihood of these conditions assessed?	VVM	109	A sensitivity analysis is included to show whether the conclusions regarding the financial/economic attractiveness is robust to reasonable variations in the critical assumptions.	OK	OK
4.4.5. To determine this, was it assessed whether it is reasonable to assume that no investment would be made at a rate of return lower than the benchmark by: b. Assessing previous investment decisions by the project participants involved, and c. Determining whether the same benchmark has been applied, or d. Determining if there are verifiable circumstances that have led to a change in the benchmark	VVM	110	In the sensitivity analysis, please give information on the previous investment decisions of the project participants and show whether these investments have been realized using VER support.	CL24	OK
4.4.6. Did the project participants rely on values from Feasibility Study Reports (FSR) that are approved by national authorities for proposed project activities?	VVM	111	In the PD version 01 it is stated that the IRR calculation for the project is based on the feasibility study prepared for the Board of the company in 2007. The board decision for the approval of this FSR was sent to the validation team.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			Also the license is revised according to this FSR hence this report is also approved by national authorities.		
4.4.7. If yes: (EB38 para.54)	VVM	111			
4.4.7.1. Has the FSR been the basis of the decision to proceed with the investment in the project, i.e. that the period of time between the finalization of the FSR and the investment decision is sufficiently short for the DOE to confirm that it is unlikely in the context of the underlying project activity that the input values would have materially changed?	VVM	111	The Feasibility Study Report dated May 2007 has been the basis of investment decision for the proposed project activity. This was verified using a board meeting report dated 14.09.2007. However it is not clear when the construction started in the table 15 under section 7 of the PD version 01 showing the schedule of the proposed project activity. Please clarify the schedule.	CL25	OK
4.4.7.2. Are the values used in the PDD and associated annexes fully consistent with the FSR? If not, was the appropriateness of the values validated?	VVM	111	The values used in the PDD and associated annexes are fully consistent with the FSR.	OK	OK
4.4.7.3. On the basis of its specific local and sectoral expertise, is confirmation provided, by cross-checking or other appropriate manner, that the input values from the FSR are valid and applicable at the time of the investment decision?	VVM	111	The values used in the FSR report are the unit values set by State Hydraulic Works and these values are applicable at the time of investment decision.	OK	OK
4.5. Barrier analysis					
4.5.1. Has barrier analysis been used to demonstrate the additionality of the proposed CDM project activity?	VVM	113	The barrier analysis is used together with investment analysis to demonstrate the additionality of the proposed project activity.	OK	OK
4.5.2. If yes, does the PDD demonstrate that the proposed CDM project activity faces barriers that:	VVM	113	The following points needs to be clarified regarding the barrier analysis:	CAR09	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
a. Prevent the implementation of this type of proposed CDM project activity? b. Do not prevent the implementation of at least one of the alternatives?			1. <u>Increases in the exchange rate:</u> Under this barrier it is stated that the project owner has obtained a loan fixed to USD and changes in this exchange rate put extra burden to the company. This can not be identified as a barrier that prevents the implementation of the proposed project activity. Any Turkish investor obtaining a loan in USD is aware that this is a risk that they may face. Also the income of the proposed project activity is also fixed to € which is also a strong currency. 2. <u>Price cap for renewable electricity:</u> In this barrier it is stated that that the renewable energy law gives a minimum purchase guarantee of 5 €¢ and caps the purchase price at 5.5 €¢. The banks consider the minimum purchase guarantee for providing the loans, therefore the project had extreme difficulties to obtain non-recourse financing from banks due to low guaranteed electricity price compared to all risks attached to the project. This project activity was able to find a loan hence the price cap on renewable electricity was not a realistic barrier for this project activity.		

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			3. <u>Market Liberalization and privatization of electricity distribution:</u> This is not a specific barrier for this type of project activities and it does not prevent the project owners from investing in renewable energy projects. 4. <u>Availability of Debt Financing and Global Recession:</u> This is not a specific barrier for this type of project activities and it does not prevent the project owners from investing in renewable energy projects. 5. <u>Technical Innovation:</u> Under this barrier it is stated that the net head of the facility is 793,26 m and when the facility is commissioned, the Uzundere-I HEPP will be the one of a kind application in terms of its operational facility in Turkey. Please provide sufficient evidence to show that this project is a unique operation. 6. <u>Planning Problems With Low Scale Maps:</u> According to the Guidance given in EB50 Report Annex 13, "the barriers that can be mitigated by additional financial means can be quantified and represented as costs and should not be identified as a barrier for implementation of project while conducting the barrier analysis", therefore the unavailability		

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			of sensitive maps can not be defined as a barrier that may prevent the implementation of the project activity. 7. <u>Harsh Geological Conditions:</u> Under this barrier it is stated that <i>"Black Sea region is known with heavy rain storms and accompanying landslides. The location of the project is in a mountainous area and the project may face landslides during and after the construction. In order to prevent, any possible landslide, the project design can be changed. For example, new approach roads can be opened or excavation slopes can be increased. All these extra work may increase the cost of the project."</i> This is a risk that may be faced for all kinds of projects in the Black Sea region of Turkey. This is not a preventive barrier. Please discuss this barrier specifically for the project activity. Also under this barrier, information about another project activity (Uzundere-II-Cataldere Power house) is given. Please clarify. 8. <u>Lack of Adequate Land Registry in the Region:</u> Under this barrier it is stated that <i>"Land registry books of Turkey had not been completed yet. Therefore, especially in rural areas,</i>		

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
			while some land slots are being used by local villagers, they seem to be idle and belonging to Turkish Treasury. Hence, the lands under the use of villagers but seems to be idle were recognized during the construction stage of the project and this led to cost increases for the project." According to the Guidance given in EB50 Report Annex 13, "the barriers that can be mitigated by additional financial means can be quantified and represented as costs and should not be identified as a barrier for implementation of project while conducting the barrier analysis". Please define how this barrier prevented the implementation of the project activity.		
4.6. Common practice analysis					
4.6.1. Is this a large-scale or first-of-its kind small-scale project activity?	VVM	117	The project activity is a large scale project.	OK	OK
4.6.2. Was common practice analysis carried out as a credibility check of the other available evidence used by the project participants to demonstrate additionality?	VVM	117	Yes, the common practice analysis is carried out as a credibility check of other available evidence used by the PPs to demonstrate additionality.	OK	OK
4.6.3. Was it assessed whether the geographical scope (e.g. defined region) of the common practice analysis is appropriate for the assessment of common practice related to the project activity's technology or industry type? (For certain technologies the relevant region for assessment will be local and for others it	VVM	118	The geographical scope is identified as the same region which is Eastern Black Sea region. Considering the terrain and climatic conditions it is reasonable to choose the Eastern Black Sea region as the geographical scope of the common	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
may be trans-national /global.)			practice analysis.		
4.6.4. Was a region other than the entire host country chosen?	VVM	118	Yes the black sea region is chosen, however after the analysis on the black sea region a wider analysis on the participation of the private sector on electricity production from hydropower plants is also given in the PD version 01.	OK	OK
4.6.5. If yes, was the explanation why this region is more appropriate assessed?	VVM	118	Please explain why Eastern Black Sea Region is chosen for comparison with similar project activities.	CL26	OK
4.6.6. Using official sources and local and industry expertise, was it determined to what extent similar and operational projects (e.g., using similar technology or practice), other than CDM project activities, and have been undertaken in the defined region?	VVM	118	In the PD version 01, it is determined to what extent similar projects are undertaken in both the defined region and in Turkey using an official source (website of Energy Market Regulatory Authority). As Turkey is a voluntary market and does not have a registry system, it is not possible to identify whether these project activities have benefited from VER support. However all of the similar project activities identified under common practice analysis have very low completion rates.	OK	OK
4.6.7. Are similar and operational projects, other than CDM project activities, already "widely observed and commonly carried out" in the defined region?	VVM	118	There are only 4 projects that are similar to the proposed project activity in the region (with a similar installed capacity). None of these projects are operational yet. They are still under construction and as mentioned above they have very low completion rates.	OK	OK
4.6.8. If yes, was it assessed whether there are essential	VVM	118	N/A	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
distinctions between the proposed CDM project activity and the other similar activities?					
5. Monitoring plan					
5.1. Is this monitoring plan based on the approved monitoring methodology applied to the proposed CDM project activity?	VVM	120	The monitoring plan is based on the "Approved consolidated baseline and monitoring methodology ACM0002"	OK	OK
5.2. Does the monitoring plan contain all necessary parameters?	VVM	121	Yes the monitoring plan contains all necessary parameters.	OK	OK
5.3. Are the monitoring arrangements described in the monitoring plan feasible within the project design?	VVM	121	The monitoring arrangements described in the monitoring plan is feasible within the project design.	OK	OK
5.4. Are the means of implementation of the monitoring plan sufficient to ensure that the emission reductions achieved by/resulting from the proposed CDM project activity can be reported ex post and verified?	VVM	121	The means of implementation of the monitoring plan is sufficient enough to ensure that the emission reductions achieved by/resulting from the proposed VCS project activity can be reported ex post and verified.	OK	OK
6. Sustainable development					
6.1. Does the CDM project activity assists Parties not included in Annex I to the Convention in achieving sustainable development?	VVM	123	The project activity is developed under VCS, hence the checklist question is N/A.	OK	OK
6.2. Does the letter of approval by the DNA of the host Party confirm the contribution of the proposed CDM project activity to the sustainable development of the host Party?	VVM	124	The project activity is developed under VCS, hence the checklist question is N/A.	OK	OK
7. Local stakeholder consultation					
7.1. Were local stakeholders (public, including individuals, groups or communities affected, of likely to be affected, by the proposed CDM project activity or actions leading to the implementation of such an	VVM	126	The project activity is developed under VCS, stakeholder consultation is not a requirement of VCS, however under section 6 of the VCS PD it is stated as	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
activity) invited by the PPs to comment on the proposed CDM project activity prior to the publication of the PDD on the UNFCCC website?			follows: <i>"The project owner used different channels to invite comments by stakeholders. Most of the meetings were held through the Mukhtar (village governors) and Council of Elderly (Ihtiyar Heyeti; the village council) of Cataldere village, and the officials of Municipality of Kaptanpasa. The project owner visited various households in the project area to comprehend their needs and to identify how the project could assist in meeting the social and economic needs of the community."</i>		
7.2. Have comments by local stakeholders that can reasonably be considered relevant for the proposed CDM project activity been invited?	VVM	127	Under section 6 of the VCS PD version 01, it is stated as follows: <i>"Although the domestic laws and regulations do not require any stakeholders' participation, both in the project planning and construction phase, the project owner managed to conduct various unofficial and unstructured stakeholder participation meetings."</i> Please clarify which domestic law is referred here and how this law applies to the project activity. Please also describe in detail how the project activity is described to the stakeholders in these meetings.	CL19	OK
7.3. Is the summary of the comments received as provided in the PDD complete?	VVM	127	A summary of the comments are given under section 6 of the VCS PD version 01.	OK	OK

CHECKLIST QUESTION	Ref.	§	COMMENTS	Draft Concl	Final Concl
7.4. Have the project participants taken due account of any comments received and described this process in the PDD?	VVM	127	How due account has been taken of comments received from the stakeholders is given under section 6 of the VCS PD version 01.	OK	OK
8. Environmental impacts					
8.1. Have the project participants submitted documentation on the analysis of the environmental impacts of the project activity?	VVM	129	The project participants have submitted a detailed EIA report both to the validation team and the Turkish Ministry of Environment and Forestry.	OK	OK
8.2. Have the project participants undertaken an analysis of environmental impacts?	VVM	130	Yes, the project participants have undertaken an EIA.	OK	OK
8.3. Does the host Party require an environmental impact assessment?	VVM	130	Yes the host party requires an EIA.	OK	OK
8.4. If yes, have the environmental impact assessment approved by local government?	VVM	130	The EIA report have been approved by the Ministry of Environment and Forestry and the project activity has an EIA positive document dated 01/04/2010. This document was also submitted to the validation team.	OK	OK

TABLE 3 LEGAL REQUIREMENTS

CHECKLIST QUESTION	Ref.	MoV*	COMMENTS	Draft Concl	Final Concl
1. Legal requirements					
1.1. Is the project activity environmentally licensed by the competent authority?	PDv01	DR	The project has obtained an "EIA positive" document dated 01.04.2010, from the Ministry of Environment and Forestry. According to Turkish Law, when the proposed project activity was first planned, for HEPPs with installed capacity below 50 MW, the project first needed to prepare a pre-EIA (Environmental Impact Assessment) report and submit this report to the Ministry of Environment and Forestry. After this report has been reviewed, the Ministry either requires a further and more detailed EIA or gives an EIA not required document to the project owners. In the case of the proposed project activity the EIA not required report was taken on 03.02.2006. After obtaining this report, and before the start of the construction, the project participants have decided to change the installed capacity as the capacity change was below 20 MW, the project was not required to perform an EIA and obtained EIA not required document from the Rize Governorship Directorate of Environment and Forestry, dated 27.01.2009. However, during this process the project participants have decided to go through a full Environmental Impact Assessment, because of the changes in the Environmental Regulation. The full EIA report and "EIA not required certificate" dated 01.04.2010 that is given based on this report by The Ministry of Environment and Forest were submitted to the validation team.	OK	OK

CHECKLIST QUESTION	Ref.	MoV*	COMMENTS	Draft Concl	Final Concl
1.2. Are the conditions of the environmental license being met?	2	DR	Conditions of the environmental permit are being met.	OK	OK
1.3 Are the conditions of the Designated National Authority being met?	2	DR	Turkey does not have a DNA.	OK	OK

TABLE 4 RESOLUTION OF CORRECTIVE ACTION AND CLARIFICATION REQUESTS

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
CAR01 - The project proponent has applied CDM approved Methodology ACM0002 version 10 and tools that are referred in ACM0002. However, ACM0002 version 10 is not the latest version of the methodology at the time of PD submission to the DOE. Please revise the PD using the latest version of the methodology.	Table 1 2.a.a, 2.d.v, 2.d.ee 2.d.ii Table 2 3.11.1	To be discussed <u>Response to Review 1:</u> The latest version of the methodology (Version 11) is used in the revised PDD. The reference link for the methodology is also revised. However, since the version revision does not include any change about the application of the methodology to hydropower plants, there has been no change in the PDD except the version number. <u>Response to Review 2:</u> The Version number is corrected. Also, the applicability conditions for hydropower are added to Section 2.2 and 3.1 of the revised VCS PD. <u>Response to Review 3:</u> The corrective action is taken in the Section 2.4	<u>Review 1:</u> The version of the methodology is not changed. <u>The corrective action request is still open.</u> <u>Review 2:</u> Under section 3.1 of the VCS PD version 03, where applicability conditions are justified it is still stated that ACM0002 version 10 is used. Also under sections 2.2 and 3.1 while justifying the applicability of the methodology, please include the applicability conditions for hydro power plants (i.e. the power density of the proposed project activity) <u>The corrective action request is still open.</u> <u>Review 3:</u> Applicability conditions are justified under section 2.2 and 3.1 of the VCS PD in accordance with the methodology. Under section 2.4 of the VCS PD

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
			version 03, where the justification for project boundary it is still stated that ACM0002 version 10 is used. <u>The corrective action request is still open.</u> Review 4: The version number of the ACM0002 was corrected as 11 in all PD. <u>The corrective action request is closed.</u>
CAR02 - Section 1.17 "List of commercially sensitive information" is missing in the PD version 01.	Table 1 2.d.t	The Section is added. Response to Review 1: Section 1.17 is added.	In the PD version 02, section 1.17 is added. It is stated that there is no commercially sensitive information under section 1.17. <u>The corrective action request is closed.</u>
CAR03 - The origin of the data is given under section 3.2 of the VCS PD version 01 in a tabular format. The source of data is also given under section 3.3 of the VCS PD version 01 within the data/parameter tables. For the parameter $EG_{\text{facility},y}$ the source of data is given as "metering device from the grid operator" under section 3.2 and as "monthly invoice" under section 3.3. In the methodology the source of data for this parameter is defined as "Project	Table 1 2.d.ff.iii 2.d.gg.iii Table 2 3.27.2.1	The source of data for the parameter $EG_{\text{facility},y}$ is corrected as "Project activity site" as suggested by the methodology. Response to Review 1: The source of data is corrected as "project activity site" as suggested by the methodology in the revised PDD. The $EG_{\text{facility},y}$ value is going to be monitored via electricity meter readings. The EF, FC and NCV values are excluded	Review 1: Under section 3.2, Table 12, the source of data is given as "project activity site". Under section 3.3 in Table 13, the source of data is defined as monthly invoice for the same parameter. Please define the source of data clearly. The monthly invoices may include transmission line losses depending on whether the project owner has bilateral agreements or not,

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
activity site” Please define the same source for the same parameter and make sure the source is in line with the methodology		from the Monitoring Plan and explanations about the Monitoring are revised accordingly. <u>Response to Review 2:</u> For the parameter Cap_{PJ}, the monitoring way is revised and the following explanation is added to the Monitoring Section of the revised VCS PD. “The second data to be monitored is the installed capacity of the project activity. The data will be monitored from the supplier information on the equipments which shows the installed capacity of the turbines. Since the installed capacity is not a dynamic value, there won’t be any continuous measurement as it is done in electricity generation measurement. The plant manager will be responsible to provide the installed capacity data.”	therefore may not be a very reliable source for monitoring the total electricity production. According to the tool monthly invoices shall be used for cross-checking the reported monitoring data. In the monitoring plan $EF_{grid,CM,y}$, FC and NCV values are also included in the monitoring plan, and the frequencies of these parameters are defined as “once for each crediting period”. As the EF is calculated ex-ante, at PPs own choice, these parameters can be excluded from the monitoring plan. <u>The corrective action request is still open.</u> <u>Review 2:</u> The data source is corrected for the parameter and EF, FC and NCV is excluded from the monitoring plan as the grid emission factor is calculated ex-ante for the first crediting period. For the parameter Cap_{PJ} it is stated that the monitoring will be done from the digital displays on the equipment. Please clarify how the value will be monitored, does the digital displays show the instantaneous value of the

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
			turbines or the total installed capacity? How will this data be recorded? Will it be recorded continuously? <u>The corrective action request is still open.</u> <u>Review 3:</u> The installed capacity of the project (Cap_{PJ}) will be monitored from the supplier information on the equipments that shows the installed capacity of the turbines. The installed capacity is monitored yearly since the parameter does not change continuously. The monitoring information of this parameter is in accordance with methodology. <u>The corrective action request is closed.</u>
CAR04 - Under section 2.4 of the PD, after identifying the baseline scenario, please also include a description of the technology that would be employed and/or the activities that would take place in the absence of the proposed VCS project activity.	Table 2 3.22	Not understood <u>Response to Review 1:</u> In the absence of the project activity, no technology would be employed and the baseline scenario would continue. The explanation below is added to the end of the Section 2.4, after the baseline emissions calculation: "In the baseline scenario, Uzundere-I HEPP	<u>Review 1:</u> Under section 2.4 of the VCS PD, after determining the baseline scenario, please give a brief description on the technology or activities that would be employed in the absence of the project activity taking into consideration the characteristics of the Turkish grid. <u>The corrective action request is still open.</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
		would not be constructed. Therefore, in the absence of the proposed VCS project activity, no other technology would be employed and the electricity amount to be generated by the project would be met by the Turkey's current national electricity grid. This would lead to 87,407 tCO ₂ e emissions per annum."	Review 2: At the end of section 2.4 of the PD version 03, a description of the activities that would take place in the absence of the project activity is given. This description is in fact the same as the baseline scenario and is accepted by the validation team. <u>The corrective action request is closed.</u>
CAR05 - The project emissions are calculated under sections 1.3 and 2.4 of the PD version 01. Please clarify and correct the following points regarding the power density calculations: 1. Under section 1.3, the value of Cap_{PJ} is given as 84,000,000 W. Under section 2.4, the value of Cap_{PJ} is given as 20,500,000 W, However the installed capacity of the project was defined as 63 MW in other sections of the PD version 01. 2. Under section 1.3, the value of A_{PJ} is given as 13.4 km², under section 2.4 of the PD the value of A_{PJ} is given as 13,837 m². Please clarify where this value is taken from and which value is the correct one by giving references. 3. Please revise the Power Density	Table 2 3.24.2.1	The inconsistencies are corrected. The values which are used in the power density calculation are corrected as Installed capacity: 63,000,000 W Reservoir are : XX.XXX m2 And the calculation is revised accordingly. Response to Review 1: (Reservoir area) A_{PJ} value is 7275.54 m² for the project. The PD calculations and the monitoring section of the PDD is revised accordingly. Also, please find attached document to verify the reservoir area of the project. (<i>INCESU REGULATORU ALAN VE HACMI.PDF</i>) Response to Review 2:	Review 1: The value of A_{PJ} is not corrected. Please clarify the value of A_{PJ} and please give reference sources for this value. Please revise the power density calculations. <u>The corrective action request is still open.</u> Review 2: In the PD version 03, only the surface area of Incesu Regulator is taken into consideration while calculating the power density of the proposed project activity. The reasoning for this choice is stated as follows: "The project has two diversion weirs: Uzundere and Incesu. Since Uzundere is a tyrol type regulator, only the

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
calculations using the correct values, making sure that the calculations and the data used in the calculations are the same throughout the whole PD.		The corrective action is taken in Section 1.3	<i>Incesu regulator area is included in the power density calculation."</i> Please clearly identify by giving reference sources why tyrol type regulators are not included in the calculation. <u>The corrective action request is still open.</u> Review 3: The pictures as an evidence of Uzundere is a tyrol type regulator is added to VCS PD. Also, it is validated from the feasibility report that Uzundere is a tyrol type regulator. The power density is not calculated for the Uzundere Regulator since the tyrol type water intake structure has not a reservoir. <u>The corrective action request is closed.</u>
CAR06 - The justifications about the selections have been given in the PD version 01. However the following issues need correction: 1. Under section 2.4, Step 2 of the PD version 01, it is stated that "option C of Simple OM method was adopted. However "tool to calculate	Table 2 3.24.5	1. The selected option for Simple OM method is corrected as "Oprtion B" in the PDD Version 02. 2. The wording of the section is revised. Please see the PDD Version 02. 3. The justification of the choice of the Option B is presented under Step 3	Review 1: 1. In the VCS PD version 02, Simple OM Option B is chosen. OK. 2. The wording of the section is revised in the PD version 02, however the share of low cost/must run resources are still calculated for an average of 5 most recent years.

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
emission factor for an electricity system" does not have any option C for the Simple OM method. Please clearly identify the method chosen under this step 2. Under section 2.4 step 2, it is further stated that "Simple OM Option C Method is eligible when low cost and/or must run resources constitute as an average of the five most recent years, less than 50% of the total generation of the grid" However in the tool this prerequisite is given for all of the options of Simple OM and it is stated as follows: <i>"The simple OM method (option a) can only be used if low-cost/must-run resources constitute less than 50% of total grid generation in: 1) average of the five most recent years, or 2) based on long-term averages for hydroelectricity production."</i> Please justify why the average of five most recent years is calculated. The tool suggests calculating the share of LCMR based on long term averages. 3. In the PD version 01 under the section 2.4 Step 4, Simple OM Option B is used however there is		of Section 2.4 of the PDD Version 02. 4. About the NCV calculation, the explanation "The fuel consumption values (FC_{i,y}) for each fossil fuel type are taken from the TEIAS statistics. Also, the net calorific values (NCV_{i,y}) are derived from these statistics by dividing total calorific values of fossil fuel types (GJ) by total amount of fossil fuel consumed in that type of power plants (tonnes)." Is added to the PDD Version 02. 5. The explanation "Since the option (a) corresponds less than 20% of the system generation, the option (b) is used for BM calculations." Is added to the PDD Version 02. 6. EF_{EL,m,y} value is not calculated in the PDD, since this value is calculated via formula (2) or formula (3) when Option A is chosen. However, in Uzundere-I PDD, Option B of Simple OM method is chosen and EF_{grid,BM,y} value is calculated in line with the tool. <u>Response to Review 1:</u> 2)- The table is revised and an explanation is added to the revised PDD. In the revised PDD, it is explained that "The share of hydroelectricity	However according to the tool, "The simple OM method (option a) can only be used if low-cost/must-run resources constitute less than 50% of total grid generation in: 1) average of the five most recent years, or 2) based on long-term averages for hydroelectricity production." As the subject project activity is a HEPP and also as LCMR resources in the Turkish Grid is mainly consisting of HEPPs, the calculation shall be made as a long term average. Please correct accordingly. 3. The use of Simple OM Option B is justified in the PD version 02. OK 4. How NCV of fuels are calculated is explained in the VCS PD version 02. OK 5. The choice of option (b) is justified under step 5 of the baseline calculations in the VCS PD version 02. OK 6. According to the tool the Build Margin Emission Factor shall be calculated using the following formula:

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
no justification regarding this choice. According to the Tool there are some applicability conditions regarding the Simple OM option B, please justify your choice taking into consideration the conditions set out in the tool. 4. In the baseline calculation excel sheet, it is seen that the value of NCV of fuels is calculated using data from TEIAS, however under section 2.4 step 4, it is not explained and justified how the NCV is calculated. 5. Under section 2.4 Step 5, there are two options namely Option 1 and Option 2, and according to the tool the option chosen shall be documented in the PD. Please document and justify your choice under Step 5. 6. Under section 2.4 Step 6 the formula $EF_{grid,BM,y}$ is given. According to the tool <i>"The CO2 emission factor of each power unit m ($EF_{EL,m,y}$) should be determined as per the guidance in Step 4 (a) for the simple OM, using options A1, A2 or A3, using for y the most recent historical year for which power generation data is</i>		production of Turkey has not exceeded 50% for 20 years (since 1989)" 6)- BM Calculation Section of the PDD is revised.	$EF_{grid,BM,y} = \frac{\sum_m EG_{m,y} \times EF_{EL,m,y}}{\sum_m EG_{m,y}}$ In this formula $EF_{EL,m,y}$ corresponds to "CO2 emission factor of power unit m in year y (tCO2/MWh)". Again according to the tool (p.16) <i>"$EF_{EL,m,y}$ should be determined as per the guidance in Step 4 (a) for the simple OM, using options A1, A2 or A3, using for y the most recent historical year for which power generation data is available, and using for m the power units included in the build margin."</i> Therefore please revise your calculations using the formulas given in the tool. <u>The corrective action request is still open.</u> Review 2: 2) In the PD version 03 the table showing the share of low cost/ must run resources are given between 2000-2008. Also it is explained that the share of HEPP production has not

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
<i>available, and using for m the power units included in the build margin."</i> However it is not clear in the PD which equation is chosen for the calculation of the $EF_{EL,m,y}$. Please clearly identify which equation is used and justify this selection.			exceeded 50% for 20 years. These statements were validated through the website of TEIAS which shows statistical values of the Turkish Grid. 6) The BM calculation section is revised in the PD version 03. The formula used to calculate EF_{EL} is given in the PD. <u>The corrective action request is closed.</u>
CAR07 - The equity IRR calculations reflect the license duration as an assessment period. However the period of assessment shall be limited to the lifetime of the turbines (operational lifetime). Please revise the assessment period taking into consideration that the turbines are manufactured by Alstom.	Table 2 4.1.13.8	The period of assessment was assumed to be 35 years. Please see "Karadeniz Uzundere I _IRR v2.0 07 June 2010" excel file <u>Response to Review 1:</u> The calculation sheet was revised. "Karadeniz Uzundere I _IRR v2.0 19 July 2010" excel file <u>Response to Review 2:</u> - The corrective action regarding with the assessment period (35 years) is taken in the PD. - The corrective action regarding with Sub-Step 2a is taken in the PD.	<u>Review 1:</u> The revised IRR calculation sheet is not submitted to the validation team. <u>The corrective action request is still open.</u> <u>Review 2:</u> The assessment period is revised in the revised IRR calculation sheet which was submitted to the validation team. However in the PD it is stated that the financial analysis is performed over 49 years. This statement is not in line with the calculations. Please revise the PD so that it reflects the correct assumptions and the correct calculations. Also under Sub-Step 2a of the investment analysis, it is stated that benchmark analysis is assessed to be

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
			appropriate, however why benchmark analysis is selected is not justified. Please justify your selection under this step. <u>The corrective action request is still open.</u> Review 3: It is stated in the VCS PD that the financial analysis is performed over the 35 year period from 2010 to 2045. The benchmark selection is justified under sub-step 2a. <u>The corrective action request is closed.</u>
CAR08 - The following issues needs to be clarified regarding the investment analysis section of the PD version 01 and the excel calculation sheet named "Karadeniz Uzundere I_IRR.xls": 1. Please be in line with the statutory accounting principles regarding the tax income. For example, please start the income tax application after the cumulative income becomes positive. 2. In the cash flow scenarios in which VER Revenue is included, there must be income tax charged for the additional income of VER. In the	Table 2 4.4.3	To be discussed in detail Response to Review 1: The corrective actions were taken as follows: 1. The tax income application was started after the cumulative income becomes positive in 2015. 2. The tax income application was started for VER income after the cumulative income becomes positive in 2015 as can be seen in the VER Revenues table in	Review 1: <u>The corrective action request is still open.</u> Review 2: 1. According to the IRR calculation sheet the cumulative income becomes positive firstly on the year 2012. Therefore the income tax should be added starting from 2012. 2. Please take 80% of VER revenue as a cash inflow in the

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
analysis, VER income is added to the original cash flow directly. Please, apply income tax for the VER incomes. 3. Please provide the finance expense figures in the analysis supported by the formulas. It seems very high; 98.8 k \$. That's why the IRR figure is very low. Please note that, the total amount received is 73.8m \$. 4. In relation with this comment, it is seen that there is an interest expense in 2009 and creates a negative cash flow at the end of year. Please clarify how this negative balance will be compensated. If it is thought that, this negative balance will be compensated by the shareholders' equity, then change the loan/equity ratio accordingly. 5. Please present whether VAT (Value Addet Tax – KDV) is included in the investment amount of 92.7m \$ or not? If it is included, it should be deducted in the following years. 6. In the analysis, it is seen that asset amortization starts before the construction period is finished. It would be better to start the amortization after the installation is fully completed based on the statutory		the revised calculation sheet "Karadeniz Uzundere I _IRR v2.0 07 June 2010" excel file 3. The mistake made in the IRR sheet was corrected. In row 96; the figure in the older version of the document included did not show the real interest expenses. The mistake was corrected. Due to this correction the IRR figures of the project have changed and the related amendments were made in the PDD. Of a formula for the finance expenses; due to the fact that the credit facility used in the project provided by a islamic bank (Bank Asya), there does not exist a standard formula to reach the figures. In practice, for each and every type of payable, Karadeniz sends a payment instruction to Bank Asya to make the payment to the related third party, and Bank Asya sends a payment table to Karadeniz which includes the principal amount and profit share of the Bank and due times based an islamic calculations. Karadeniz accumulates these documents	analysis. Because, 20% of the revenue should be paid as a tax. 3. 31,6mio \$ is calculated as an interest expense for a 92,7mio \$ loan. Please provide more detailed specifications of loan so that the results can be assessed by the validation team. 4. OK 5. OK 6. OK 7. The depreciation periods are revised in the revised IRR calculations, however this revision is not reflected to the PD. Please revise the PD so that it is in line with the calculation sheet. 8. OK 9. OK 10. Total asset value is 87m\$ and the total depreciated amount is 84,9m\$. The left amount should be add as a cash inflow at the end of the analysis period as the fair value of the assets. 11. Please include your response in the validation protocol. <u>The corrective action request is still open.</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
accounting principles. 7. Please correct the depreciation periods. Construction must be 40 years and electromechanical equipment must be 30 years. 8. Please verify that there is no license fee in the investment. 9. Please verify if the Debt Repayments represented in the IRR analysis are the Capital Repayments. If it is so, the total debt repayment amount (50.1m \$) is not equal to the Debt figure (73.7m \$) in row 124 in the analysis. Please correct it. 10. Please correct the depreciation periods at first, then include the fair value of the project as a cash inflow at the end of the assessment period. 11. In the PDD 10% of contingency is added to the Project investment but in the IRR analysis it is not taken into consideration. Please be in line with the PDD.		to calculate its debt to the Bank. Karadeniz does not have a granda table for its whole outstanding debts to Bank Asya provided by Bank Asya. 4. The inequality is stemming from a mistake made in the IRR sheet. 100% of the investment was financed by the credit facility provided by Bank Asya. 5. VAT is not included in the investment amount. 6. The amortization was started at 2010. 7. The suggested periods were considered. 8. There is no license fee in the investment. 9. The inequality is stemming from a mistake made in the IRR sheet. 100% of the investment was financed by the credit facility provided by Bank Asya. 10. Since the end of the cashflow schedule is very close to the ecd of the licensing period, the fair value of the project is negligible. <u>Response to Review 2:</u>	<u>Review 3:</u> 1. Please note that the cumulative net profit in 2011 is negative so that there must not a corporate tax. Please start the corporate application when the cumulative net profit first becomes positive. 2. OK 3. OK 4. OK 5. OK 6. OK 7. OK 8. OK 9. OK 10. There should be tax applied to the scrap value. Please take the 80% of the scrap value. Please clarify. 11. The responses are stated in the validation protocol. OK <u>The corrective action request is still open.</u> <u>Review 3:</u> 1. The first tax payment is moved to 2012 when the cumulative profit is positive for the first

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
		1. The income tax was started from 2011. 2. 80% of VER revenue was assumed as cash inflow in the analysis. 3. Of a formula for the finance expenses; due to the fact that the credit facility used in the project provided by a islamic bank (Bank Asya), there does not exist a standard formula to reach the figures. In practice, for each and every type of payable, Karadeniz sends a payment instruction to Bank Asya to make the payment to the related third party, and Bank Asya sends a payment table to Karadeniz which includes the principal amount and profit share of the Bank and due times based an islamic calculations. Karadeniz accumulates these documents to calculate its debt to the Bank. Karadeniz does not have a granda table for its whole outstanding debts to Bank Asya provided by Bank Asya. Please see the attached excel sheet (Loan Calculation Table) 4. OK 5. OK	- time. OK 2. OK 3. OK 4. OK 5. OK 6. OK 7. OK 8. OK 9. OK 10. 20% of income tax is applied to the scrap value. OK 11. OK <u>The corrective action request is closed.</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
		6. OK 7. The depreciation periods were revised in the PD. 8. OK 9. OK 10. The depreciation equipment was stopped at the end of 30th year. The difference of 6.6 m\$ between total asset value (87.8m\$) and the total depreciated amount (81,2m\$) was added as a cash inflow at the end of the analysis period as the fair value of the assets (cell no AN113 in the IRR calculation sheet). 11. Was not understood <u>Response to Review 3:</u> 1. The corrective action is taken for the year 2011. The tax application is started for the year 2010 based on cumulative profit. 2. OK 3. OK 4. OK 5. OK 6. OK 7. OK 8. OK	

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
		9. OK 10. the 80% of the scrap value is taken. 11. OK In order to reflect the changes occurred, the Project Cashflow Tables are renewed in the PDD (Table 10) and Sensitivity Analysis.	
CAR09 - The following points needs to be clarified regarding the barrier analysis: 1. <u>Increases in the exchange rate:</u> Under this barrier it is stated that the project owner has obtained a loan fixed to USD and changes in this exchange rate put extra burden to the company. This can not be identified as a barrier that prevents the implementation of the proposed project activity. Any Turkish investor obtaining a loan in USD is aware that this is a risk that they may face. Also the income of the proposed project activity is also fixed to € which is also a strong currency. 2. <u>Price cap for renewable electricity:</u> In this barrier it is stated that that the renewable energy law gives a minimum purchase guarantee of 5 €¢	Table 2 4.5.2	To be added after the 2 nd Review <u>Response to Review 2:</u> 1. OK 2. The Bank that provided the loan to the project has considered the potential VERs of the project assessing the loan application. Please find the attached Bank Letter (Bank Letter September 2009) 3. ... 4. Although this is not a specific barrier for this type of project activities, it is a barriers for Uzundere_1 HEPP. The project owner had experienced many difficulties in finding financing options to realize the project. As can be seen in IRR sheet, the	<u>Review 1:</u> <u>The corrective action request is still open.</u> <u>Review 2:</u> <u>The corrective action request is still open.</u> <u>Review 3:</u> 1. <u>Increases in the exchange rate:</u> There is no change in the PD regarding the corrective action request. Please evaluate the corrective action request. 2. <u>Price cap for renewable electricity:</u> The bank letter was submitted to the validation team. The letter shows that the VER revenue is a major positive effect for lender to provide loan. According to the

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
and caps the purchase price at 5.5 €¢. The banks consider the minimum purchase guarantee for providing the loans, therefore the project had extreme difficulties to obtain non-recourse financing from banks due to low guaranteed electricity price compared to all risks attached to the project. This project activity was able to find a loan hence the price cap on renewable electricity was not a realistic barrier for this project activity. 3. <u>Market Liberalization and privatization of electricity distribution:</u> This is not a specific barrier for this type of project activities and it does not prevent the project owners from investing in renewable energy projects. 4. <u>Availability of Debt Financing and Global Recession:</u> This is not a specific barrier for this type of project activities and it does not prevent the project owners from investing in renewable energy projects. 5. <u>Technical Innovation:</u> Under this barrier it is stated that the net head of the facility is 793,26 m and when the facility is commissioned, the		interest burden regarding with the debt facility is relatively high. As a result of this challenge, the Bank made it clear that they consider the importance of potential VER revenue in the letter in the attachment (Bank Letter September 2009) 5. Please see the attached the excel sheet (The Head Load Information for the Projects Licensed by EMRA), and the explanation added to the revised PD. 6. Besides the fact that the lack of detailed maps might lead to additional and unforeseeable costs in construction period, that challenge leads to delays in time schedule due to geological formations and conditions which can not be foreseen on low scale maps. 7. It is not true that the barrier of harsh geological conditions is a risk for all kinds of projects in the Black Sea region of Turkey. For instance, in case a project owner contemplates to erect a thermal power facility, it can	Guidance given in EB50 Report Annex 13, "for the cases where the investment is done by a company which also purchases the CERs and the loan agreement mentions that, there is an objective demonstration that the CDM facilitated the lending.", therefore the bank letter is accepted as objective evidence. 3. <u>Market Liberalization and privatization of electricity distribution:</u> There is no change in the PD regarding the corrective action request. Please evaluate the corrective action request. 4. <u>Availability of Debt Financing and Global Recession:</u> Price cap for renewable electricity is identified an investment barrier because of the project had experienced extreme difficulty in getting non-recourse financing from banks. It has justified that the VER revenue is a major positive effect for lender to provide loan. However, a similar barrier, <i>Availability of Debt Financing and Global Recession</i>, is identified as an investment barrier. As a result, difficulty of debt financing is considered two times as an investment barrier. Please

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
Uzundere-I HEPP will be the one of a kind application in terms of its operational facility in Turkey. Please provide sufficient evidence to show that this project is a unique operation. 6. <u>Planning Problems With Low Scale Maps:</u> According to the Guidance given in EB50 Report Annex 13, "the barriers that can be mitigated by additional financial means can be quantified and represented as costs and should not be identified as a barrier for implementation of project while conducting the barrier analysis", therefore the unavailability of sensitive maps can not be defined as a barrier that may prevent the implementation of the project activity. 7. <u>Harsh Geological Conditions:</u> Under this barrier it is stated that <i>"Black Sea region is known with heavy rain storms and accompanying landslides. The location of the project is in a mountainous area and the project may face landslides during and after the construction. In order to prevent, any possible landslide, the project design can be changed. For example, new approach roads can be opened or excavation slopes can be increased. All these extra work may increase the</i>		easily erect such a facility on a plain geographical part of the region. But Karadeniz had to build a HEPP facility on a mountainous area to use the head load. The information given about another project activity (Uzundere-II-Cataldere Power house) is typo whgich was taken from another PD. It was deleted. 8. This barrier does not only lead to cost increase. The lack of adequate land registries also led additional bureaucratic processes and so that the construction of the project has faced delays. <u>Response to Review 3:</u> 1. OK. The argument is deleted form PD. 2. OK 3. OK. The argument is deleted form PD. 4. OK. The argument is deleted from PD. Some of the arguments	clarify. 5. <u>Technical Innovation:</u> It is stated that the explanation added to the revised PD. However, there is no extra explanation in the PD. The reference given regarding head load is checked. Please clearly explain why this statement is considered as technical innovation. 6. <u>Planning Problems With Low Scale Maps:</u> This problem is known at the beginning of the project activity. The unavailability of sensitive maps can not be defined as a barrier that may prevent the implementation of the project activity. Please clarify. 7. <u>Harsh Geological Conditions:</u> Harsh geological conditions of Black Sea region is a risk that may be faced for all kinds of projects in the Black Sea region of Turkey. Also lots of HEPP is set up at the mountainous area and it is known at the feasibility studies. Please justify that VER revenue is a major effect to overcome harsh geological conditions. 8. <u>Lack of Adequate Land Registry in the Region:</u> This is not a specific barrier for this project. This

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
<i>cost of the project."</i> This is a risk that may be faced for all kinds of projects in the Black Sea region of Turkey. This is not a preventive barrier. Please discuss this barrier specifically for the project activity. Also under this barrier, information about another project activity (Uzundere-II-Cataldere Power house) is given. Please clarify. 8. <u>Lack of Adequate Land Registry in the Region:</u> Under this barrier it is stated that <i>"Land registry books of Turkey had not been completed yet. Therefore, especially in rural areas, while some land slots are being used by local villagers, they seem to be idle and belonging to Turkish Treasury. Hence, the lands under the use of villagers but seems to be idle were recognized during the construction stage of the project and this led to cost increases for the project."</i> According to the Guidance given in EB50 Report Annex 13, "the barriers that can be mitigated by additional financial means can be quantified and represented as costs and should not be identified as a barrier for implementation of project while		for that barrier is added to Price Cap for Renewable Electricity in order to strengthen the argument 5. The corrective action is taken 6. Regarding with the additionality, the comparison should be made between the proposed activity and the alternatives stated in Section 2.5 Sub-step 1a instead of a comparison that would be made between same types of project activities in the same geographical region. For instance, possible planning problems with low scale maps would constitute a much less risk, and unforeseeable costs in case the investor had preferred to establish a thermal (natural gas) power plant which would have been erected relatively small area with the same annual power output instead of HEPP in the Black Sea Region. Such a comparison would reveal and prove the justification for this barrier. The explanation is given in the PD. 7. Regarding with the additionality,	problem is known at the beginning of the project activity. The lack of adequate land registry in the region can not be defined as a barrier that may prevent the implementation of the project activity. Please clarify. <u>The corrective action request is still open.</u> Review 4: According to the <i>"Tool for the demonstration and assessment of additionality"</i>, if barrier analysis is used, determine whether the proposed project activity faces barriers that: (a) Prevent the implementation of this type of proposed project activity; and (b) Do not prevent the implementation of at least one of the alternatives. The identified barriers are only sufficient grounds for demonstration of additionality if they would prevent potential project proponents from carrying out the proposed project activity undertaken without being registered as a CDM project activity. If the CDM does not alleviate the identified barriers that prevent the proposed project activity from

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
conducting the barrier analysis". Please define how this barrier prevented the implementation of the project activity.		the comparison should be made between the proposed activity and the alternatives stated in Section 2.5 Sub-step 1a instead of a comparison that would be made between same types of project activities in the same geographical region. For instance, harsh geological conditions would constitute a much less risk, and unforeseeable costs in case the investor had preferred to establish a thermal (natural gas) power plant which would have been erected relatively small area with the same annual power output instead of HEPP in the Black Sea Region. Such a comparison would reveal and prove the justification for this barrier. The explanation is given in the PD. 8. Regarding with the additionality, the comparison should be made between the proposed activity and the alternatives stated in Section 2.5 Sub-step 1a instead of a comparison that would be made between same types of project activities in the same	occurring, then the project activity is not additional. Please explain and demonstrate your barriers in line with tool and Guidelines for Objective Demonstration and assessment of Barriers (EB50, Annex 13). <u>The corrective action request is still open.</u> Review 5: The barrier analysis was excluded from the PD version 04. The assessment and demonstration of additionality of the project is done by investment analysis. <u>The corrective action request is closed.</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
		geographical region. For instance, lack of adequate land registry would constitute a much less risk, and unforeseeable costs in case the investor had preferred to establish a thermal (natural gas) power plant which would have been erected relatively small area with the same annual power output instead of HEPP in the Black Sea Region. Such a comparison would reveal and prove the justification for this barrier. The explanation is given in the PD. <u>Response to Review 4:</u> The corrective action is taken in the PD.	
CAR 10 - The efficiency computations are based on total power generation data. However, the Build Margin Option B refers to <i>"the set of capacity additions in the electricity system that comprise 20% of the system generation and that have been built most recently"</i>. The efficiency of new plants is typically higher than old ones. However the		The efficiency factors are taken from the Annex I of the tool in the revised PDD. Please see related changes in the attached Baseline Calculation Excel Sheet and the PDD Version 03.	In the PD version 03, the baseline calculations are revised using the default efficiency factors given in Annex I of the tool. <u>The corrective action request is closed.</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
calculations include both the new plants and the old plants that are operational in the years of calculation (i.e. all of the thermal power plants supplying the Turkish grid) Therefore, this approach does not result in a conservative estimation of the efficiencies and may lead to higher figures in emission reductions. Please use the default efficiency values for new units (built after 2000) given in the TOOL be employed.			
CL01 - The detailed information on the project location, including geographic and physical information allowing the unique identification and delineation of the specific extent of the project are provided under section 1.5 of the VCS PD, however the following points needs further clarification: 1. Please provide the GPS coordinates separately for the power house and the reservoirs. 2. Please also identify the grid connection point for the power plant. 3. Please clarify what the second arrow represents in Figure 2. "Project Location on Turkey Map"	Table 1 2.d.f Table 2 3.6.2	Detailed GPS coordinates for the points 1 and 2 was added to PDD as a table. The second arrow mentioned in the point 3 which was mistakenly drawn was deleted	The detailed coordinates of the project activity is given in the VCS PD version 02. <u>The clarification request is closed.</u>
CL02 - Under Section 1.6 of the VCS PD version 01, the following information is	Table 1	To be added after the 2nd Review	<u>Review 1:</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
given; ⌘ The starting date of the project activity: July 1, 2010 when the plant commences electricity generation. ⌘ The date on which project reached financial closure: November 2006. Please also state the date on which a financial commitment was made to the project activity.	2.d.g.i	The point was not understood. The explanation if necessary will be provided later	<u>The clarification request is still open.</u> Review 2: <u>The clarification request is still open.</u> Review 3: The information stated in the section 1.6 of the VCS PD is found satisfactory. <u>The clarification request is closed.</u>
CL03 - The project technologies are given under section 1.9 of the VCS PD. Also information on the products, services and expected level of activity is given under the same section. Under this section it is stated that the turbines will be supplied from Alstom, and it is also stated that the expected lifetime of the equipment is around 20 years. Please clarify the above mentioned statements as the lifetime of the Alstom turbines are usually longer than 20 years.	Table 1 2.d.j	The clarification was made. Response to Review 1: The corrective action was taken in both PDD and the investment analysis.	Review 1: Under section 1.9 of the VCS PD version 02, the statement is revised as "<i>The first-hand turbines which will be used in the project will be supplied from Alstom of which the lifetime are usually longer than 20 years.</i>" Please define the lifetime of the turbines clearly. Over 20 years does not give a view of the projected lifetime, it can be 25 or 55 years, which are totally different. Please make sure the statement is in line with the technical lifetime assumptions made for the investment analysis. <u>The clarification request is still open.</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
			Review 2: In the PD version 03 the projected lifetime of the turbines are given as 35 years, this assumption is also in line with the IRR calculations. <u>The clarification request is closed.</u>
CL04 - In VCS PD version 01, the following risks are identified: ⊗ decrease of rainfall in the region due to the climate change ⊗ Technical risks: The risks that stem from misoptimization of the installed power from water flow as well as choosing the not best-fit turbines and technical equipment for the design. Please clarify how the identified technical risk has a potential to effect projects' GHG emission reductions. The installed power of the project activity has already been changed from 43.68 MW to 63.0 MW and the project activity uses Alstom Turbines, which are optimized taking into consideration the flow regime.	Table 1 2.d.I	To be added after the 2nd Review The point was not understood. The explanation if necessary will be provided later <u>Response to Review 3</u> OK. The corrective action is taken in the PD.	Review 1: <u>The clarification request is still open.</u> Review 2: <u>The clarification request is still open.</u> Review 3: The installed power of the project activity has already changed 43.68 MW to 63.0 MW. Therefore, misoptimization of the installed power can not be defined as a technical risk. Please clarify. <u>The clarification request is still open.</u> Review 4: The technical risk was excluded from the PD.

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
			<u>The clarification request is closed.</u>
CL05 - The contact information of the project proponent and carbon consultants are given in a tabular format under section 1.15 of the VCS PD. Please also briefly define the roles and responsibilities of the contact persons regarding the carbon crediting process.	Table 1 2.d.r	To be added <u>Response to Review 1:</u> The clarification was provided under section 1.15	<u>Review 1:</u> <u>The clarification request is still open.</u> <u>Review 2:</u> The roles and responsibilities of the contact persons both for the PP and the carbon consultants are given under section 1.15 of the PD version 03. <u>The clarification request is closed.</u>
CL06 - Under section 2.4 of the PD version 01, the GHG sources are identified for both the baseline scenario and the project activity. However in the table CH₄ emissions from the reservoir has been considered twice. For the operation and construction of the HEPP CH₄ emissions are considered to be negligible, but it is stated on the next row that CH₄ emissions from reservoir is a main emission source. Also under this section the same information with the exact same sentences are given twice. Please clarify.	Table 1 2.d.v Table 2 3.8.4 3.13	The table under Section 2.4 is corrected and the repeated paragraphs are deleted. Please see the PDD Version 02.	In the VCS PD version 02, the emission sources are explained in detail. The repeated information is also taken out of the PD. <u>The clarification request is closed.</u>
CL07 - Types of data and information to be reported including the units of	Table 1 2.d.ff.ii	Corrected as "Total amount of fossil fuel consumed by power plants". Please	<u>Review 1:</u> The description of FC_{i,y} is revised in the

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
measurement is given under section 3.2 of the VCS PD version 01 in a tabular format; however the parameter $FC_{i,y}$ is defined as: "<i>Turkey's gross electricity generated by primary energy source of the five most recent years</i>" Please clarify.		see the revised table in the PDD Version 02. <u>Response to Review 1:</u> The $FC_{i,y}$, $NCV_{i,y}$, and $EF_{grid,CM,y}$ values are excluded from the Monitoring Plan.	VCS PD version 02. According to the methodology ACM0002, as the Grid Emission Factor is calculated ex-ante for the first crediting period, the parameters $FC_{i,y}$, $NCV_{i,y}$, and $EF_{grid,CM,y}$ only needs to be calculated and/or monitored at the beginning of each crediting period. In the PD, the monitoring frequencies of these parameters are defined as 'once for each crediting period' which is in line with the methodology. However as these parameters will not be monitored for the first crediting period, they can be excluded from the monitoring plan. Please confirm whether these parameters will be included in the MP or not. <u>The clarification request is still open.</u> <u>Review 2:</u> In the PDD version 03, as the combined margin grid emission factor is calculated ex-ante, the parameters FC, NCV and EF are excluded from the monitoring plan. This exclusion is in line with the monitoring methodology. <u>The clarification request is closed.</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
CL08 - In the VCS PD version 01 the value applied for the parameter CAP_{PJ} is given as 20.5 MW, however it is stated in other sections of the PD that the installed capacity of the project activity is 63 MW. Please clarify. Also for the parameter A_{PJ} the value is given as 13,837 m². This parameter was given as 13.4 km² in some other section of the PD. Please clarify the Reservoir Area and please also define where this value is taken from. Please include all of your references in the PD.	Table 1 2.d.gg.iv	<u>Response to Review 1:</u> One of the regulators of the project is tyrol type (Uzundere). Therefore, there is not a reservoir area to be used in power density calculation for this regulator. However, the other regulator (Incesu regulator) is included in the power density calculation. The reservoir area of the Incesu regulator is measured as 7,275.54 m². Please see the revised power density calculations in the PDD Version 03. Also, you can find attached document to verify the reservoir area. <u>Response to Review 2:</u> The corrective action is taken in Section 1.3 <u>Response to Review 3:</u> The area of 43.05 km² in the Feasibility Study Report is the rainfall area, not the reservoir area. The reservoir area of the Incesu regulator is justified with a scheme (Picture 7) in the Section 1.3	<u>Review 1:</u> During the site visit it was seen that there are two regulators in the proposed project activity, please include the surface area of both of the regulators in the Power Density calculations. <u>The clarification request is still open.</u> <u>Review 2:</u> Please give more detailed information about tyrol type regulators and justify why there is not a reservoir area. Please give references to justify your assumptions. <u>The clarification request is still open.</u> <u>Review 3:</u> The pictures as an evidence of Uzundere is a tyrol type regulator is added to VCS PD. Also, it is validated from the feasibility report that Uzundere is a tyrol type regulator. The power density is not calculated for the Uzundere Regulator since the tyrol type water intake structure has not a reservoir. The regulator area is given as 43.05

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
			km² in the Feasibility Study Report dated May 2007. Please justify the reservoir area of the Incesu regulator taken as 7,275.54 m². <u>The corrective action request is still open.</u> <u>Review 4:</u> The technical drawing of the Incesu Regulator is provided in the VCS PD version 04 as a evidence of the reservoir area. <u>The corrective action request is closed.</u>
CL09 - Measurement procedures for each data/parameter is given in the tabular format, however for the parameter EG_{facility,y} the measurement procedure is given as MWh. Please define the measurement procedures as per the methodology.	Table 1 2.d.gg.v Table 2 3.27.2.2	The measurement procedure for parameter EG_{facility,y} is corrected as "Electricity meter" in the PDD Version 02. <u>Response to Review 1:</u> The electricity production figure of the power plant will not be monitored via monthly bills. The electricity production will be monitored via electricity meter readings. The PDD is revised accordingly. Besides, the explanations requested by validation team are aded to the PDD Version 03.	<u>Review 1:</u> The measurement procedure is revised as electricity meter in the PD version 02. In the monitoring plan please clearly identify how the EG_{facility,y} will be monitored. In section 3.2 of the PD version 02, it is stated that the electricity generation figure will be taken from the bills. Please clearly identify how many meters will be installed?, what kind of bills the electricity generation figures will be taken from?. Please also clearly identify the calibration procedures, roles and

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
			responsibilities regarding the monitoring, how will the data be collected and how it will be cross-checked? <u>The clarification request is still open.</u> <u>Review 2:</u> In the PD version 03 the monitoring plan is revised. The electricity production will be monitored via electricity meter readings. Calibration procedures and roles and responsibilities are also given under section 3.2 of the PDD. <u>The clarification request is closed.</u>
CL10 - Please also include the version of the "Tool to calculate the emission factor for an electricity system"	Table 1 2.d.ii	The version of the tool (Version 02) is mentioned together with the methodology in the PDD Version 02.	The tool version is included throughout the document in the PD version 02. <u>The clarification request is closed.</u>
CL11 - A summary of the environmental impacts is given under section 5 of the VCS PD version 01. However in this section it is stated that the domestic law and regulation does not require any environmental impact assessment. Please clarify as this is not the case at the moment. Please also include a summary of the EIA	Table 1 2.d.mm	During the licensing of the project as 43,68 MW, it was not required to have a EIA. Later because the increase demand for installed capacity is lower than 25 MW since the difference 63,0 MW and 43,68 MW is 19,32, it was not required then. <u>Response to Review 1:</u>	<u>Review 1:</u> The project has gone through an environmental impact analysis and has prepared an EIA report in 2010. A copy of this report was submitted to the validation team. Please include a summary of this EIA report in the PD, and please revise the EIA section of the PD with the latest

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
prepared by the project owner which was also sent to the validation team.		A summary of the EIA was included (Section 6), and EIA Section (Section 6) was revised accordingly.	available information. <u>The clarification request is still open.</u> Review 2: Under section 5 of the PD version 03, a summary of the EIA is given. <u>The clarification request is closed.</u>
CL12 - Under section 1.4 of the VCS PD version 01 a brief description of the project activity is given covering the purpose of the project activity. Please also describe the scenario existing prior to the start of the project, project scenario and the baseline scenario under this section of the PD.	Table 2 3.4.1	To be discussed in detail Response to Review 1: The scenario existing prior to the project activity and the baseline scenario is added under Section 1.4 of the PDD Version 03.	Review 1: <u>The clarification request is still open.</u> Review 2: Under section 1.4 of the PD version 03, the scenario existing prior to the start of the proposed project activity is given. <u>The clarification request is closed.</u>
CL13 - Under section 1.4 of the VCS PD please give a brief description on how the proposed project activity reduces greenhouse gas emissions.	Table 2 3.4.3	To be discussed in detail Response to Review 1: "The project will generate electricity from hydropower and will result in emission reductions (relatively when compared to the baseline scenario) in parallel with its electricity generation figures." Please see Section 1.4 of the PDD Version 03.	Review 1: <u>The clarification request is still open.</u> Review 2: The explanation is added under section 1.4 of the PD version 03. <u>The clarification request is closed.</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
CL14 - The equipment that will be used is given under section 1.9 of the PD version 01. Please give a more detailed list and arrangement of the main manufacturing / production technologies, systems and equipments involved in the proposed project activity.	Table 2 3.8.3	Detailed information was added under section 1.9	Detailed information about the project technologies is given under section 1.9 of the VCS PD version 02. <u>The clarification request is closed.</u>
CL15 – 1. Under section 2.4 of the PD version 01, Baseline emissions formula is given as: $BE_y = (EG_y - EG_{baseline}) \times EF_{Grid,CM,y}$ However this is not the formula given in the methodology for Greenfield power plants. Although the results will not change, please revise the PD so that the correct formulas are included in the PD. Also under the same section, the value of EG_y is given as 100,646 MWh/year however in the baseline calculations, the grid emission factor is multiplied by 156,205 MWh/year. Please clarify. 2. Under section 1.3 of the PD version 01, although the power density is calculated to be greater than 10 W/m ² the Project Emission formula is given after this calculation (the formula is given twice under this section) Also it is stated that the	Table 2 3.24.2	1. The formula is corrected in line with the tool and the wording of section is revised in the PDD Version 02. 2. The repeated formula and the wording mistake about $PE_{HP,y}$ are deleted from the PDD Version 02.	The formula and the wording is corrected in the PD version 02. <u>The clarification request is closed.</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
annual $PE_{HP,y}$ is given in Table 1 which does not exist in this section. Please clarify.			
CL16 - The equations and parameters used in the PD are correct in general however under the section 2.4 Step 4, the formula of the $EF_{grid,OMsimple,y}$ is given however the explanation of this formula does not match the explanation given in the tool. Please revise the PD to include the correct explanations.	Table 2 3.24.6	The explanation under the formula of the Step 4 is corrected in the PDD Version 02.	In the PD version 03 the explanation of the formula is revised to be in line with the tool. <u>The clarification request is closed.</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
CL17 - Please give more detailed information on how each equation is applied to enable the reader to reproduce the calculations. For example, under section 2.4, step 7: For the calculation of the $EF_{grid,CM,y}$ when the values given in the PD are used in the formula, the result is 0,6135 however this value is taken as 0,614 in the calculations. Under the same section for the calculation of BE_y when the values given are used in the formula the result is 95,909.87 tCO_{2e}, however the result is given as 95,927 tCO_{2e} in the PD version 01. Please clarify.	Table 2 3.24.7	<u>Response to Review 1:</u> The calculated values are not rounded up in the revised PDD. Also, the BM calculation is revised in the PDD Version 03. <u>Response to Review 2:</u> The baseline emissions value is not rounded up. Since the EF value is given in the PDD by being rounded down, this value is seen as if it is more than the exact value. However, the OM, BM and combined EF values are given in 5 decimals. The one who reads the VCS PD, can verify the baseline emissions value by seeing the Annex 3 of the VCS PD. A footnote (footnote 37) is added under Step 7 of the Baseline Calculation, to inform the reader about the values are given in the VCS PD by being rounded down and the exact values are given under Annex 3. In addition, these values are also given in 7 decimals in the Baseline Calculation Excel Sheet.	<u>Review 1:</u> <u>The clarification request is still open.</u> <u>Review 2:</u> In the PDD under section 2.4 the annual emission reductions are calculated as follows: "$BE_y = 156,205 \times 0.559 = 87,407 \text{ tCO}_2\text{e}$" When the above calculation is made the resulting baseline emissions shall be 87.318 tons of CO₂. This value is still rounded up to 87,407. Please either change this value or show all of the decimal places so that the calculations are correct. In the excel calculation sheet, under baseline EF tab, please also show all of the decimal places if they are going to be used in the PD. <u>The clarification request is still open.</u> <u>Review 3:</u> The information on how each equation is applied to enable the reader to reproduce the calculations is stated in the PD.

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
			<u>The clarification request is closed.</u>
CL18 - Under section 2.1 of the VCS PD version 01, it is stated that the completion date of the baseline study is 06/10/2009, however the date of the first version of the PD is March 16 2010. Please clarify.	Table 2 3.31.1	The mistakenly written expression was corrected	The completion date of the baseline study is revised as 16.03.2010 in the VCS PD version 02. <u>The clarification request is closed.</u>
CL19 - Under section 6 of the VCS PD version 01, it is stated as follows: <i>"Although the domestic laws and regulations do not require any stakeholders' participation, both in the project planning and construction phase, the project owner managed to conduct various unofficial and unstructured stakeholder participation meetings."</i> Please clarify which domestic law is referred here and how this law applies to the project activity. Please also describe in detail how the project activity is described to the stakeholders in these meetings.	Table 2 3.34.2 7.2	To be discussed in detail <u>Response to Review 1:</u> The corrective action was taken in section 6. <u>Response to Review 2:</u> - The related domestic laws are Regulation on Environmental Impact Assessment (Date: 16.12.2003, Number: 25318), and Regulation on Environmental Impact Assessment (Date: 17.07.2008, Number: 26939). According to the Article 7 of both regulation the stakeholders' participation meetings must be held for the projects which are required to have a EIA: - The information about the Stakeholder Participation Meeting was held on 14.04.2009 at 13.00 in Cukurluhoca Village Coffeehouse,	<u>Review 1:</u> <u>The clarification request is still open.</u> <u>Review 2:</u> Under section 6 of the PD version 03 brief explanation on how the project activity is described to the stakeholders is given, however it is not explained which domestic law is referred in the 1st paragraph. It is also not explained how the referred law applies to the proposed project activity. Also under section 6 it is stated that the stakeholder consultation meetings were finalized in 2007. However under section 5 of the PD version 03 it is stated as follows: <i>"Within the scope of the EIA of the project, a Stakeholder Participation Meeting was held on 14.04.2009 at 13.00 in Cukurluhoca Village</i>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
		Cayeli, Rize was added in Section 6. - How the project activity is described to the stakeholders in these meetings was mentioned in the Section 6. <u>Response to Review 3:</u> - The related domestic laws are added to VCS PD (Annex 6 and Annex 7). - The meetings which were finalized in November 2007 were the ones held before the Rize Administrative Court ordered the preparation of an EIA. Upon the order of the Court, the second round of stakeholder consultation meeting within the scope of EIA was held in 14th April 2009. <u>Response to Review 4:</u> The corrective action is taken on the PD. The Annexes 6&7 were deleted.	<i>Coffeehouse, Cayeli, Rize."</i> Please also include information about this stakeholder consultation meeting in the PD. <u>The clarification request is still open.</u> <u>Review 3:</u> - The related domestic laws are explained in the protocol. Please also add the related domestic laws to VCS PD. - Stakeholder Participation Meeting was held on 14.04.2009. However, it is stated in the VCS PD, page 53, that the stakeholder consultation meetings were finalized in 2007. Please clearly explain whether two stakeholder meetings were held or not in order to prevent any misunderstandings. - It is explained in section 6 of the VCS PD how the project activity is described to the stakeholders. <u>The clarification request is still open.</u> <u>Review 4:</u> - Please only add the information regarding the laws to section 6.. - Two stakeholder meetings were held.

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
			The meetings that were finalized in November 2007 were the ones held before the Rize Administrative Court ordered the preparation of an EIA. The second round of stakeholder consultation meeting within the scope of EIA was held in 14th April 2009. <u>The clarification request is still open.</u> <u>Review 5:</u> The information regarding the laws is stated in to section 6 of the VCS PD version 04 and the Annexes 6 and 7 was removed from the PD. <u>The clarification request is closed.</u>
CL20 - Please clarify when the stakeholder consultation process is finalized.	Table 2 3.34.3	The exact place of the needed amendment in the PDD is not clear <u>Response to Review 1:</u> The corrective action was taken in section 6. <u>Response to Review 2:</u> The corrective action was taken in section 6. <u>Response to Review 3:</u>	<u>Review 1:</u> Under Section '6. Stakeholder's Comments' please clarify when the stakeholder consultation process was finalized. <u>The clarification request is still open.</u> <u>Review 2:</u> Under section 6 of the PD version 03 it is stated that the stakeholder consultation meetings were finalized in November 2007. Please also give

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
		- The meetings which were finalized in November 2007 were the ones held before the Rize Administrative Court ordered the preparation of an EIA as stated in the VCS PD, page 53. Upon the order of the Court, the second round of stakeholder consultation meeting within the scope of EIA was held in 14th April 2009.	information about the second round of stakeholder consultation meeting which was realized within the scope of EIA and when it is finalized. <u>The clarification request is still open.</u> <u>Review 3:</u> - Stakeholder Participation Meeting was held on 14.04.2009. However, it is stated in the VCS PD, page 53, that the stakeholder consultation meetings were finalized in 2007. Please clearly explain whether two stakeholder meetings were held or not in order to prevent any misunderstandings. - It is stated in the PD that the second round of stakeholder consultation meeting within the scope of EIA which was ordered to be prepared by Rize Administrative Court was held in 14.04.2009. However, it understood from the statements in VCS PD that first round was held on 14.04.2009. Please clearly explain. <u>The clarification request is still open.</u> <u>Review 4:</u> Two stakeholder meetings were held.

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
			The meetings that were finalized in November 2007 were the ones held before the Rize Administrative Court ordered the preparation of an EIA. The second round of stakeholder consultation meeting within the scope of EIA was held in 14th April 2009. <u>The clarification request is closed.</u>
CL21 - Please identify the names of the stakeholders that have made comments.	Table 2 3.35.1	The exact place of the needed amendment in the PDD is not clear <u>Response to Review 1:</u> The corrective action was taken in section 6.	<u>Review 1:</u> Under Section '6. Stakeholder's Comments' please identify the names of the stakeholders that have made comments. <u>The clarification request is still open.</u> <u>Review 2:</u> The names of the stakeholders that have made comments are given in the PD version 03. <u>The clarification request is closed.</u>
CL22 - The Benchmark Analysis (Option III) is selected however it is not justified why benchmark analysis is selected. Please clarify.	Table 2 4.1.9.2	To be added after the 2nd Review <u>Response to Review 1:</u> The explanation was provided in the PD at Sub-Step 2a	<u>Review 1:</u> <u>The clarification request is still open.</u> <u>Review 2:</u> <u>The clarification request is still open.</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
			<u>Review 3:</u> It is stated in the VCS PD that the financial analysis is performed over the 35 year period from 2010 to 2045. The benchmark selection is justified under sub-step 2a. <u>The clarification request is closed.</u>
CL23 - The benchmark selected is given as follows under Sub-Step 2b: <i>"Almost all bankers who have been asked for an opinion for the minimum IRR that is required for the approval of a similar project expressed some after tax IRR that is in the bandwidth of 16 to 18 percent."</i> This type of benchmark is suitable for the type of IRR calculated, however please send objective evidence to prove this statement. Please be informed that at any stage of this validation, the proof of this benchmark can be requested to be made public from the PP.	Table 2 4.1.12.1	To be added after the 2nd Review <u>Response to Review 1&2:</u> Due to the difficulty stemming from finding publicly available information on bankers' view, the IRR used in previous version of the PDD was deleted. Instead, the calculated benchmark IRR was provided in the revised PD, and that IRR was used as benchmark.	<u>Review 1:</u> <u>The clarification request is still open.</u> <u>Review 2:</u> <u>The clarification request is still open.</u> <u>Review 3:</u> The calculated benchmark IRR was provided in the revised PD. <u>The clarification request is closed.</u>
CL24 - In the sensitivity analysis, please give information on the previous investment decisions of the project participants and show whether these investments have been realized using VER support.	Table 2 4.4.5	To be added after the 3rd Review <u>Response to Review 3:</u> As a result of the correspondence with BV Verifier Ms. Isil Timuroglu, it was	<u>Review 1:</u> <u>The clarification request is still open.</u> <u>Review 2:</u>

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
		decided that a sensitivity analysis regarding with the previous investment decision is not necessary at this phase of the process.	<u>The clarification request is still open.</u> Review 3: <u>The clarification request is still open.</u> Review 4: Benchmark calculation is calculated by the CAPM model in a correct way. Therefore, it does not need to perform further analysis. <u>The clarification request is closed.</u>
CL25 - The Feasibility Study Report dated May 2007 has been the basis of investment decision for the proposed project activity. This was verified using a board meeting report dated 14.09.2007. However it is not clear when the construction started in the table 15 under section 7 of the PD version 01 showing the schedule of the proposed project activity. Please clarify the schedule.	Table 2 4.4.7.1	To be added after the 2nd Review Response to Review 1&2: As shown in the table 15 the the first civil works were started for the project at 1st quarter of 2007 based on the first design choice for the project for 43 MW, and in May 2007 a revised feasibility report for increased installed capacity was prepared and the regarding decision was made by the board of the Company before the construction works had not gone too far.	Review 1: <u>The clarification request is still open.</u> Review 2: <u>The clarification request is still open.</u> Review 3: The explanation is accepted by the validation team. <u>The clarification request is closed.</u>
CL26 - Please explain why Eastern Black Sea Region is chosen for comparison with	Table 2 4.6.5	To be added after the 2 nd Review	Review 1:

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
similar project activities.		<u>Response to Review 1:</u> Since the project is located in the Eastern Black Sea Region, and each geographical region in Turkey has its own peculiar conditions which might impact the projects, this region was chosen for comparison. <u>Response to Review 3:</u> The corrective action is taken in the table 16 and_Sub-step 4a: Other activities similar to the proposed project activity section of PD.	<u>The clarification request is still open.</u> <u>Review 2:</u> <u>The clarification request is still open.</u> <u>Review 3:</u> 1. Please give the references of the each value given in Table 16. 2. Please add the date for completion rate. 3. Please add the completion rate of the proposed project. 4. Other similar activities in the region are not stated in the Table 16. Please explain the limit of installed capacity is considered as similar project. 5. It is stated in the Step 4 that all these projects are run-off-river type hydro projects. Please justify. <u>The clarification request is still open.</u> <u>Review 4:</u> 1. The references of the each value given in Table 16 was provided in the VCS PD version 04. 2. The date of completion rate, May

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
			2010, was added to the VCS PD version 04. 3. The completion rate (100%) of the proposed project was added to the VCS PD version 04. 4. The similar projects is defined as the installed capacity of ranging between 53 MW and 73 MW, -10 MW and +10 MW of the installed capacity of Uzundere 1 HEPP in Eastern Black Sea Region. 5. The type of the hydro projects was added to the Table 16 and the reference was provided. <u>The clarification request is closed.</u>
CL27 – Please clearly identify the project activities' EIA process. According to the recent news, there is a trial regarding the EIA of the proposed project activity, and the court has granted a motion for stay of execution of the project activity. Please clarify.		<u>Response to Review 1:</u> The corrective action was taken in section 5 and 6. <u>Response to Review 2:</u> Upon the positive EIA report dated 01.04.2010, a local association applied to court against the report. And the court granted a motion for stay of execution of the project activity. However, that association has withdrawn its application to court as a result of the negotiations with the project owner company. The court has	<u>Review 1:</u> <u>The clarification request is still open.</u> <u>Review 2:</u> EIA process given in section 5 and 6 is about EIA decision document dated 01.04.2010. However, the date of the news is 30.04.2010 that is a trial regarding the EIA of the proposed project activity, and the court has granted a motion for stay of execution of the project activity. Please clarify.

Draft report clarifications and corrective action requests by validation team	Ref. to checklist question in tables	Summary of project owner response	Validation team conclusion
		annulled its decision in August 2010.	<u>The clarification request is still open.</u> Review 3: The court decision dated 18/08/2010 was provided to the validation team. <u>The clarification request is closed.</u>
CL28 – Please apply depreciation for the unexpected costs.		The depreciation calculation was corrected.	The depreciation was applied to the unexpected cost. <u>The clarification request is closed.</u>